

ΠΟΛΥΤΕΧΝΕΙΟ ΚΡΗΤΗΣ

Β' ΦΑΣΗ ΕΡΓΑΣΤΗΡΙΑΚΗΣ ΕΡΓΑΣΙΑΣ



**Μία αναφορά για το μάθημα
ΒΑΣΕΙΣ ΔΕΔΟΜΕΝΩΝ – ΠΛΗ 303**

**ΓΙΑΟΥΠΙ ΚΛΕΪΝΤΑ ΑΜ : 2019030132
ΣΤΑΥΡΟΠΟΥΛΟΥ ΕΛΕΝΗ ΑΜ : 2019030070**

Χανιά, 2022

Παραδοτέο από την Α΄ Φάση

3. Ανάκτηση δεδομένων και υπολογισμοί (υλοποίηση με χρήση συναρτήσεων PostgreSQL)

3.8. (*)

- **q_3_8_get_diplomas()**: εύρεση όλων των τίτλων διπλωματικών εργασιών που όλα τα μέλη της επιτροπής εργάζονται στο ίδιο εργαστήριο.

3.9. (*)

- **q_3_9_get_courses(character)**: Ανάκτηση όλων των μαθημάτων που είναι προαπαιτούμενα ή συνιστώμενα, άμεσα ή έμμεσα, για ένα συγκεκριμένο μάθημα του οποίου δίνεται ο κωδικός.

6. Λειτουργικότητα με χρήση όψεων (views)

6.2. (*)

- **q_6_2_get_num_of_students**: Ανάκτηση του πλήθους των φοιτητών ανά έτος εγγραφής για τα τελευταία 10 έτη οι οποίοι ικανοποιούν τις προϋποθέσεις αποφοίτησης και δεν έχουν ακόμη εκπονήσει διπλωματική εργασία.

Παραδοτέο από την Β΄ Φάση

1. Επικοινωνία Java εφαρμογής με το σύστημα διαχείρισης βάσεων δεδομένων PostgreSQL μέσω JDBC:

Το eclipse project δημιουργήθηκε για να τρέχει στην βάση που επισυνάπτεται.

2. Μελέτη απόδοσης ερωτήσεων – φυσικός σχεδιασμός:

Βήμα 1:

Αρχικά, χρησιμοποιώντας το αρχείο ασφαλείας (backup) της βάσης δεδομένων που συνοδεύει την εκφώνηση το ερώτημα που μελετήθηκε γράφεται ως εξής:

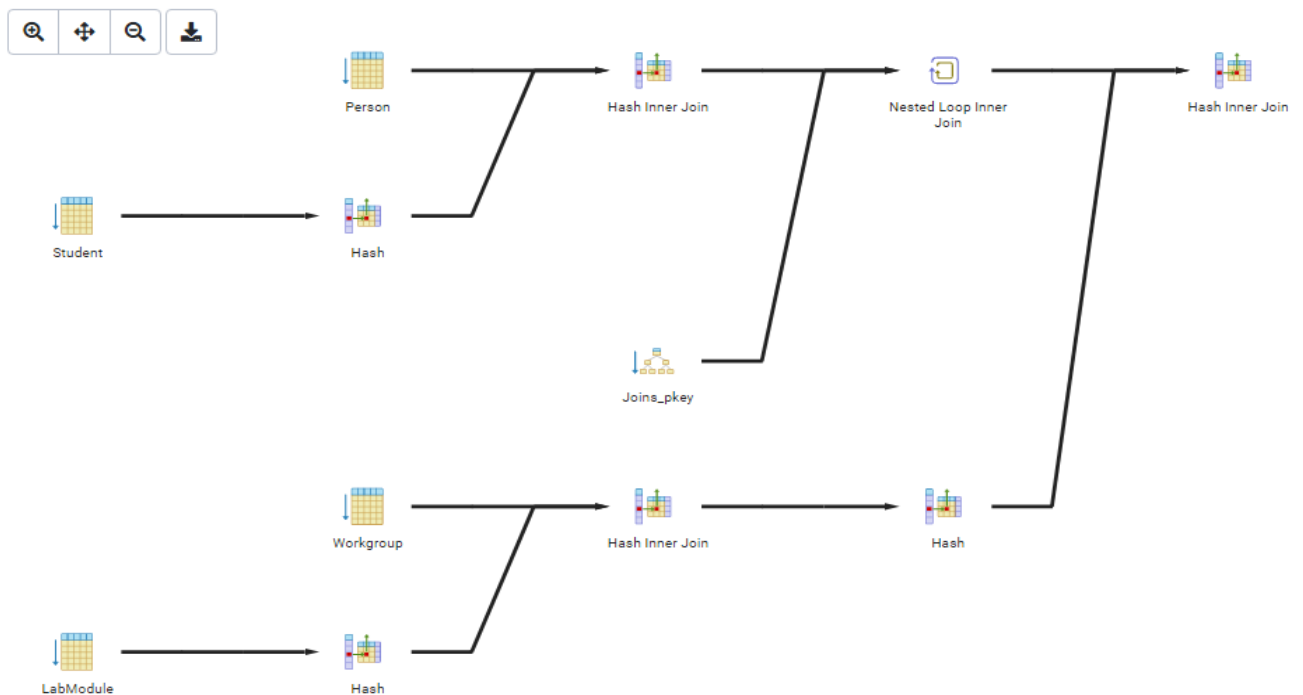
```
SELECT name, surname
FROM "Person" p JOIN "Student" s ON (p.amka = s.amka)
JOIN "Joins" j ON (j.amka = s.amka)
JOIN "Workgroup" w ON (w."wgID" = j."wgID")
JOIN "LabModule" l ON (l.module_no = w.module_no) AND (l.module_no = j.module_no)
WHERE s.entry_date > '2010-09-01'::date
AND s.entry_date < '2011-09-01'::date
AND l.type = 'project'
AND w.grade > 8
```

Βήμα 2:

Έπειτα, αφού απενεργοποιήθηκε η δυνατότητα δημιουργίας παράλληλων πλάνων εκτέλεσης, εκτελέστηκε η εντολή Explain Analyse. Ενδεικτικά πάρθηκαν 3 χρόνοι εκτέλεσης και εξήχθη ο μέσος όρος αυτών:

ΧΩΡΙΣ ΕΥΡΕΤΗΡΙΑ	
Εκτελέσεις	Χρόνος Απόκρισης (ms)
1 ^η	168.500
2 ^η	172.082
3 ^η	177.148
ΜΟ	172.577

Γραφικά, το πλάνο εκτέλεσης ήταν το εξής:



Πιο αναλυτικά, γίνεται sequential scan και inner join μεταξύ των πινάκων. Τα στιγμιότυπα από το πλάνο εκτέλεσης για 3 ενδεικτικές εκτελέσεις φαίνονται παρακάτω:

	QUERY PLAN	
	text	
1	Hash Join (cost=8784.67..20135.42 rows=1 width=34) (actual time=78.537..168.333 rows=11 loops=1)	
2	[...] Hash Cond: ((j."wglD" = w."wglD") AND (j.module_no = l.module_no))	
3	[...] -> Nested Loop (cost=1263.06..11674.61 rows=125225 width=42) (actual time=6.493..95.505 rows=124772 loops=1)	
4	[...] -> Hash Join (cost=1262.63..2763.84 rows=4532 width=58) (actual time=6.439..23.815 rows=4551 loops=1)	
5	[...] Hash Cond: ((p.amka)::text = (s.amka)::text)	
6	[...] -> Seq Scan on "Person" p (cost=0.00..1354.78 rows=55778 width=46) (actual time=0.012..3.737 rows=55778 loops=1)	
7	[...] -> Hash (cost=1205.98..1205.98 rows=4532 width=12) (actual time=6.386..6.388 rows=4551 loops=1)	
8	[...] Buckets: 8192 Batches: 1 Memory Usage: 260kB	
9	[...] -> Seq Scan on "Student" s (cost=0.00..1205.98 rows=4532 width=12) (actual time=0.024..5.787 rows=4551 loops=1)	
10	[...] Filter: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))	
11	[...] Rows Removed by Filter: 49381	
12	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..1.69 rows=28 width=20) (actual time=0.010..0.013 rows=27 loops=4551)	
13	[...] Index Cond: (amka = (p.amka)::text)	
14	[...] Heap Fetches: 0	

15	[...] -> Hash (cost=7521.11..7521.11 rows=33 width=12) (actual time=64.195..64.199 rows=32 loops=1)
16	[...] Buckets: 1024 Batches: 1 Memory Usage: 10kB
17	[...] -> Hash Join (cost=49.71..7521.11 rows=33 width=12) (actual time=0.330..64.179 rows=32 loops=1)
18	[...] Hash Cond: (w.module_no = l.module_no)
19	[...] -> Seq Scan on "Workgroup" w (cost=0.00..7470.40 rows=179 width=8) (actual time=0.049..63.820 rows=209 loops=1)
20	[...] Filter: (grade > '8'::numeric)
21	[...] Rows Removed by Filter: 376063
22	[...] -> Hash (cost=45.40..45.40 rows=345 width=4) (actual time=0.272..0.273 rows=345 loops=1)
23	[...] Buckets: 1024 Batches: 1 Memory Usage: 21kB
24	[...] -> Seq Scan on "LabModule" l (cost=0.00..45.40 rows=345 width=4) (actual time=0.011..0.232 rows=345 loops=1)
25	[...] Filter: (type = 'project'::labmodule_type)
26	[...] Rows Removed by Filter: 1527
27	Planning Time: 1.355 ms
28	Execution Time: 168.500 ms

	QUERY PLAN	
	text	
1	Hash Join (cost=8784.67..20135.42 rows=1 width=34) (actual time=80.451..171.913 rows=11 loops=1)	
2	[...] Hash Cond: ((j."wglD" = w."wglD") AND (j.module_no = l.module_no))	
3	[...] -> Nested Loop (cost=1263.06..11674.61 rows=125225 width=42) (actual time=7.098..97.719 rows=124772 loops=1)	
4	[...] -> Hash Join (cost=1262.63..2763.84 rows=4532 width=58) (actual time=7.008..25.143 rows=4551 loops=1)	
5	[...] Hash Cond: ((p.amka)::text = (s.amka)::text)	
6	[...] -> Seq Scan on "Person" p (cost=0.00..1354.78 rows=55778 width=46) (actual time=0.026..3.779 rows=55778 loops=1)	
7	[...] -> Hash (cost=1205.98..1205.98 rows=4532 width=12) (actual time=6.930..6.931 rows=4551 loops=1)	
8	[...] Buckets: 8192 Batches: 1 Memory Usage: 260kB	
9	[...] -> Seq Scan on "Student" s (cost=0.00..1205.98 rows=4532 width=12) (actual time=0.029..6.285 rows=4551 loops=1)	
10	[...] Filter: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))	
11	[...] Rows Removed by Filter: 49381	
12	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..1.69 rows=28 width=20) (actual time=0.011..0.014 rows=27 loops=4551)	
13	[...] Index Cond: (amka = (p.amka)::text)	
14	[...] Heap Fetches: 0	

15	[...] -> Hash (cost=7521.11..7521.11 rows=33 width=12) (actual time=65.549..65.552 rows=32 loops=1)
16	[...] Buckets: 1024 Batches: 1 Memory Usage: 10kB
17	[...] -> Hash Join (cost=49.71..7521.11 rows=33 width=12) (actual time=0.364..65.504 rows=32 loops=1)
18	[...] Hash Cond: (w.module_no = l.module_no)
19	[...] -> Seq Scan on "Workgroup" w (cost=0.00..7470.40 rows=179 width=8) (actual time=0.064..65.112 rows=209 loops=1)
20	[...] Filter: (grade > '8'::numeric)
21	[...] Rows Removed by Filter: 376063
22	[...] -> Hash (cost=45.40..45.40 rows=345 width=4) (actual time=0.289..0.291 rows=345 loops=1)
23	[...] Buckets: 1024 Batches: 1 Memory Usage: 21kB
24	[...] -> Seq Scan on "LabModule" l (cost=0.00..45.40 rows=345 width=4) (actual time=0.013..0.244 rows=345 loops=1)
25	[...] Filter: (type = 'project'::labmodule_type)
26	[...] Rows Removed by Filter: 1527
27	Planning Time: 1.432 ms
28	Execution Time: 172.082 ms

	QUERY PLAN	
	text	
1	Hash Join (cost=8784.67..20135.42 rows=1 width=34) (actual time=82.952..176.939 rows=11 loops=1)	
2	[...] Hash Cond: ((j."wgID" = w."wgID") AND (j.module_no = l.module_no))	
3	[...] -> Nested Loop (cost=1263.06..11674.61 rows=125225 width=42) (actual time=6.801..100.261 rows=124772 loops=1)	
4	[...] -> Hash Join (cost=1262.63..2763.84 rows=4532 width=58) (actual time=6.733..25.143 rows=4551 loops=1)	
5	[...] Hash Cond: ((p.amka)::text = (s.amka)::text)	
6	[...] -> Seq Scan on "Person" p (cost=0.00..1354.78 rows=55778 width=46) (actual time=0.044..3.923 rows=55778 loops=1)	
7	[...] -> Hash (cost=1205.98..1205.98 rows=4532 width=12) (actual time=6.634..6.635 rows=4551 loops=1)	
8	[...] Buckets: 8192 Batches: 1 Memory Usage: 260kB	
9	[...] -> Seq Scan on "Student" s (cost=0.00..1205.98 rows=4532 width=12) (actual time=0.034..6.002 rows=4551 loops=1)	
10	[...] Filter: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))	
11	[...] Rows Removed by Filter: 49381	
12	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..1.69 rows=28 width=20) (actual time=0.011..0.014 rows=27 loops=4551)	
13	[...] Index Cond: (amka = (p.amka)::text)	
14	[...] Heap Fetches: 0	

15	[...] -> Hash (cost=7521.11..7521.11 rows=33 width=12) (actual time=67.958..67.961 rows=32 loops=1)
16	[...] Buckets: 1024 Batches: 1 Memory Usage: 10kB
17	[...] -> Hash Join (cost=49.71..7521.11 rows=33 width=12) (actual time=0.440..67.895 rows=32 loops=1)
18	[...] Hash Cond: (w.module_no = l.module_no)
19	[...] -> Seq Scan on "Workgroup" w (cost=0.00..7470.40 rows=179 width=8) (actual time=0.059..67.413 rows=209 loops=1)
20	[...] Filter: (grade > '8'::numeric)
21	[...] Rows Removed by Filter: 376063
22	[...] -> Hash (cost=45.40..45.40 rows=345 width=4) (actual time=0.369..0.370 rows=345 loops=1)
23	[...] Buckets: 1024 Batches: 1 Memory Usage: 21kB
24	[...] -> Seq Scan on "LabModule" l (cost=0.00..45.40 rows=345 width=4) (actual time=0.011..0.325 rows=345 loops=1)
25	[...] Filter: (type = 'project'::labmodule_type)
26	[...] Rows Removed by Filter: 1527
27	Planning Time: 1.983 ms
28	Execution Time: 177.148 ms

Εν συνεχεία, μετά από εκτελέσεις με χρήση διαφόρων ευρετηρίων καταλήξαμε ότι αυτά που βελτιώνουν περισσότερο την απόδοση είναι τα εξής: btree για τις τιμές grade και entry_date, αφού εμφανίζονται σε συνθήκες ανισότητας και hash για την τιμή type, που εμφανίζεται σε ισότητα. Οι εντολές που χρησιμοποιήθηκαν ήταν οι CREATE INDEX και DROP INDEX για την κατασκευή και την διαγραφή του κάθε ευρετηρίου αντίστοιχα.

ΜΕ ΕΥΡΕΤΗΡΙΟ ΤΥΠΟΥ BTREE (grade)	
Εκτελέσεις	Χρόνος Απόκρισης (ms)
1 ^η	116.973
2 ^η	110.445
3 ^η	112.462
ΜΟ	113.293

Παρατηρούμε ότι με τη χρήση του ευρετηρίου Grade_idx υπάρχει αισθητή μείωση του μέσου χρόνου εκτέλεσης περίπου κατά 34%, γεγονός που συνάδει με τη θεωρία καθώς η συνθήκη που αφορά στον βαθμό έχει μεγάλο selectivity.

	QUERY PLAN text	
1	Hash Join (cost=1434.48..13303.83 rows=1 width=34) (actual time=14.863..116.743 rows=11 loops=1)	
2	[...] Hash Cond: ((j."wglD" = w."wglD") AND (j.module_no = l.module_no))	
3	[...] -> Nested Loop (cost=1265.30..12158.35 rows=130171 width=42) (actual time=6.366..107.154 rows=124772 loops=1)	
4	[...] -> Hash Join (cost=1264.87..2766.08 rows=4711 width=58) (actual time=6.295..27.887 rows=4551 loops=1)	
5	[...] Hash Cond: ((p.amka)::text = (s.amka)::text)	
6	[...] -> Seq Scan on "Person" p (cost=0.00..1354.78 rows=55778 width=46) (actual time=0.010..3.827 rows=55778 loops=1)	
7	[...] -> Hash (cost=1205.98..1205.98 rows=4711 width=12) (actual time=6.240..6.241 rows=4551 loops=1)	
8	[...] Buckets: 8192 Batches: 1 Memory Usage: 260kB	
9	[...] -> Seq Scan on "Student" s (cost=0.00..1205.98 rows=4711 width=12) (actual time=0.025..5.638 rows=4551 loops=1)	
10	[...] Filter: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))	
11	[...] Rows Removed by Filter: 49381	
12	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..1.70 rows=29 width=20) (actual time=0.012..0.015 rows=27 loops=4551)	
13	[...] Index Cond: (amka = (p.amka)::text)	
14	[...] Heap Fetches: 0	

15	[...] -> Hash (cost=169.10..169.10 rows=6 width=12) (actual time=0.697..0.700 rows=32 loops=1)
16	[...] Buckets: 1024 Batches: 1 Memory Usage: 10kB
17	[...] -> Hash Join (cost=54.38..169.10 rows=6 width=12) (actual time=0.390..0.693 rows=32 loops=1)
18	[...] Hash Cond: (w.module_no = l.module_no)
19	[...] -> Bitmap Heap Scan on "Workgroup" w (cost=4.66..119.21 rows=31 width=8) (actual time=0.066..0.336 rows=209 loops=1)
20	[...] Recheck Cond: (grade > '8'::numeric)
21	[...] Heap Blocks: exact=205
22	[...] -> Bitmap Index Scan on grade_idx (cost=0.00..4.66 rows=31 width=0) (actual time=0.043..0.043 rows=209 loops=1)
23	[...] Index Cond: (grade > '8'::numeric)
24	[...] -> Hash (cost=45.40..45.40 rows=345 width=4) (actual time=0.313..0.314 rows=345 loops=1)
25	[...] Buckets: 1024 Batches: 1 Memory Usage: 21kB
26	[...] -> Seq Scan on "LabModule" l (cost=0.00..45.40 rows=345 width=4) (actual time=0.014..0.268 rows=345 loops=1)
27	[...] Filter: (type = 'project'::labmodule_type)
28	[...] Rows Removed by Filter: 1527
29	Planning Time: 1.443 ms
30	Execution Time: 116.973 ms

	QUERY PLAN	
	text	
1	Hash Join (cost=1434.48..13303.83 rows=1 width=34) (actual time=14.861..110.211 rows=11 loops=1)	
2	[...] Hash Cond: ((j."wglD" = w."wglD") AND (j.module_no = l.module_no))	
3	[...] -> Nested Loop (cost=1265.30..12158.35 rows=130171 width=42) (actual time=6.216..100.775 rows=124772 loops=1)	
4	[...] -> Hash Join (cost=1264.87..2766.08 rows=4711 width=58) (actual time=6.151..25.633 rows=4551 loops=1)	
5	[...] Hash Cond: ((p.amka)::text = (s.amka)::text)	
6	[...] -> Seq Scan on "Person" p (cost=0.00..1354.78 rows=55778 width=46) (actual time=0.011..3.949 rows=55778 loops=1)	
7	[...] -> Hash (cost=1205.98..1205.98 rows=4711 width=12) (actual time=6.090..6.091 rows=4551 loops=1)	
8	[...] Buckets: 8192 Batches: 1 Memory Usage: 260kB	
9	[...] -> Seq Scan on "Student" s (cost=0.00..1205.98 rows=4711 width=12) (actual time=0.024..5.475 rows=4551 loops=1)	
10	[...] Filter: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))	
11	[...] Rows Removed by Filter: 49381	
12	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..1.70 rows=29 width=20) (actual time=0.011..0.014 rows=27 loops=4551)	
13	[...] Index Cond: (amka = (p.amka)::text)	
14	[...] Heap Fetches: 0	
15	[...] -> Hash (cost=169.10..169.10 rows=6 width=12) (actual time=0.751..0.755 rows=32 loops=1)	
16	[...] Buckets: 1024 Batches: 1 Memory Usage: 10kB	
17	[...] -> Hash Join (cost=54.38..169.10 rows=6 width=12) (actual time=0.345..0.747 rows=32 loops=1)	
18	[...] Hash Cond: (w.module_no = l.module_no)	
19	[...] -> Bitmap Heap Scan on "Workgroup" w (cost=4.66..119.21 rows=31 width=8) (actual time=0.068..0.437 rows=209 loops=1)	
20	[...] Recheck Cond: (grade > '8'::numeric)	
21	[...] Heap Blocks: exact=205	
22	[...] -> Bitmap Index Scan on grade_idx (cost=0.00..4.66 rows=31 width=0) (actual time=0.043..0.043 rows=209 loops=1)	
23	[...] Index Cond: (grade > '8'::numeric)	
24	[...] -> Hash (cost=45.40..45.40 rows=345 width=4) (actual time=0.267..0.269 rows=345 loops=1)	
25	[...] Buckets: 1024 Batches: 1 Memory Usage: 21kB	
26	[...] -> Seq Scan on "LabModule" l (cost=0.00..45.40 rows=345 width=4) (actual time=0.014..0.223 rows=345 loops=1)	
27	[...] Filter: (type = 'project'::labmodule_type)	
28	[...] Rows Removed by Filter: 1527	
29	Planning Time: 1.278 ms	
30	Execution Time: 110.445 ms	

	QUERY PLAN	
	text	
1	Hash Join (cost=1434.48..13303.83 rows=1 width=34) (actual time=14.517..112.232 rows=11 loops=1)	
2	[...] Hash Cond: ((j."wglD" = w."wglD") AND (j.module_no = l.module_no))	
3	[...] -> Nested Loop (cost=1265.30..12158.35 rows=130171 width=42) (actual time=6.216..102.838 rows=124772 loops=1)	
4	[...] -> Hash Join (cost=1264.87..2766.08 rows=4711 width=58) (actual time=6.135..26.507 rows=4551 loops=1)	
5	[...] Hash Cond: ((p.amka)::text = (s.amka)::text)	
6	[...] -> Seq Scan on "Person" p (cost=0.00..1354.78 rows=55778 width=46) (actual time=0.010..3.669 rows=55778 loops=1)	
7	[...] -> Hash (cost=1205.98..1205.98 rows=4711 width=12) (actual time=6.082..6.083 rows=4551 loops=1)	
8	[...] Buckets: 8192 Batches: 1 Memory Usage: 260kB	
9	[...] -> Seq Scan on "Student" s (cost=0.00..1205.98 rows=4711 width=12) (actual time=0.027..5.500 rows=4551 loops=1)	
10	[...] Filter: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))	
11	[...] Rows Removed by Filter: 49381	
12	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..1.70 rows=29 width=20) (actual time=0.011..0.014 rows=27 loops=4551)	
13	[...] Index Cond: (amka = (p.amka)::text)	
14	[...] Heap Fetches: 0	
15	[...] -> Hash (cost=169.10..169.10 rows=6 width=12) (actual time=0.735..0.739 rows=32 loops=1)	
16	[...] Buckets: 1024 Batches: 1 Memory Usage: 10kB	
17	[...] -> Hash Join (cost=54.38..169.10 rows=6 width=12) (actual time=0.418..0.730 rows=32 loops=1)	
18	[...] Hash Cond: (w.module_no = l.module_no)	
19	[...] -> Bitmap Heap Scan on "Workgroup" w (cost=4.66..119.21 rows=31 width=8) (actual time=0.077..0.355 rows=209 loops=1)	
20	[...] Recheck Cond: (grade > '8'::numeric)	
21	[...] Heap Blocks: exact=205	
22	[...] -> Bitmap Index Scan on grade_idx (cost=0.00..4.66 rows=31 width=0) (actual time=0.051..0.052 rows=209 loops=1)	
23	[...] Index Cond: (grade > '8'::numeric)	
24	[...] -> Hash (cost=45.40..45.40 rows=345 width=4) (actual time=0.333..0.333 rows=345 loops=1)	
25	[...] Buckets: 1024 Batches: 1 Memory Usage: 21kB	
26	[...] -> Seq Scan on "LabModule" l (cost=0.00..45.40 rows=345 width=4) (actual time=0.015..0.287 rows=345 loops=1)	
27	[...] Filter: (type = 'project'::labmodule_type)	
28	[...] Rows Removed by Filter: 1527	
29	Planning Time: 1.516 ms	
30	Execution Time: 112.462 ms	

ΜΕ ΕΥΡΕΤΗΡΙΟ ΤΥΠΟΥ BTREE (entry_date)	
Εκτελέσεις	Χρόνος Απόκρισης (ms)
1 ^η	168.906
2 ^η	166.120
3 ^η	161.504
ΜΟ	165.510

Συμπεραίνουμε ότι με τη χρήση του ευρετηρίου Date_idx υπάρχει μικρή μείωση του μέσου χρόνου εκτέλεσης περίπου κατά 4%, γεγονός που εν μέρει συμβαδίζει με τη θεωρία καθώς η συνθήκη που αφορά στην ημερομηνία δεν έχει τόσο μεγάλο selectivity. Ωστόσο, παίζει ρόλο και το γεγονός ότι ο αριθμός των πλειάδων στο backup αρχείο που δόθηκε δεν είναι τόσο μεγάλος για να υπάρχει ξεκάθαρη βελτίωση του χρόνου.

	QUERY PLAN	
	text	
1	Hash Join (cost=7733.70..19499.58 rows=1 width=34) (actual time=69.251..168.685 rows=11 loops=1)	
2	[...] Hash Cond: ((j."wglD" = w."wglD") AND (j.module_no = l.module_no))	
3	[...] -> Nested Loop (cost=212.22..10999.31 rows=130503 width=42) (actual time=1.696..100.321 rows=124772 loops=1)	
4	[...] -> Hash Join (cost=211.79..1713.00 rows=4723 width=58) (actual time=1.641..23.108 rows=4551 loops=1)	
5	[...] Hash Cond: ((p.amka)::text = (s.amka)::text)	
6	[...] -> Seq Scan on "Person" p (cost=0.00..1354.78 rows=55778 width=46) (actual time=0.048..3.778 rows=55778 loops=1)	
7	[...] -> Hash (cost=152.75..152.75 rows=4723 width=12) (actual time=1.507..1.509 rows=4551 loops=1)	
8	[...] Buckets: 8192 Batches: 1 Memory Usage: 260kB	
9	[...] -> Index Scan using date_idx on "Student" s (cost=0.29..152.75 rows=4723 width=12) (actual time=0.045..0.816 rows=4551 loops=1)	
10	[...] Index Cond: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))	
11	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..1.69 rows=28 width=20) (actual time=0.012..0.015 rows=27 loops=4551)	
12	[...] Index Cond: (amka = (p.amka)::text)	
13	[...] Heap Fetches: 0	
14	[...] -> Hash (cost=7521.03..7521.03 rows=30 width=12) (actual time=59.596..59.599 rows=32 loops=1)	

15	[...] Buckets: 1024 Batches: 1 Memory Usage: 10kB
16	[...] -> Hash Join (cost=49.71..7521.03 rows=30 width=12) (actual time=0.426..59.535 rows=32 loops=1)
17	[...] Hash Cond: (w.module_no = l.module_no)
18	[...] -> Seq Scan on "Workgroup" w (cost=0.00..7470.40 rows=165 width=8) (actual time=0.062..59.073 rows=209 loops=1)
19	[...] Filter: (grade > '8'::numeric)
20	[...] Rows Removed by Filter: 376063
21	[...] -> Hash (cost=45.40..45.40 rows=345 width=4) (actual time=0.342..0.343 rows=345 loops=1)
22	[...] Buckets: 1024 Batches: 1 Memory Usage: 21kB
23	[...] -> Seq Scan on "LabModule" l (cost=0.00..45.40 rows=345 width=4) (actual time=0.012..0.284 rows=345 loops=1)
24	[...] Filter: (type = 'project'::labmodule_type)
25	[...] Rows Removed by Filter: 1527
26	Planning Time: 2.201 ms
27	Execution Time: 168.906 ms

	QUERY PLAN	
	text	
1	Hash Join (cost=7733.70..19499.58 rows=1 width=34) (actual time=70.865..165.929 rows=11 loops=1)	
2	[...] Hash Cond: ((j."wglD" = w."wglD") AND (j.module_no = l.module_no))	
3	[...] -> Nested Loop (cost=212.22..10999.31 rows=130503 width=42) (actual time=1.700..95.891 rows=124772 loops=1)	
4	[...] -> Hash Join (cost=211.79..1713.00 rows=4723 width=58) (actual time=1.661..20.764 rows=4551 loops=1)	
5	[...] Hash Cond: ((p.amka)::text = (s.amka)::text)	
6	[...] -> Seq Scan on "Person" p (cost=0.00..1354.78 rows=55778 width=46) (actual time=0.030..3.666 rows=55778 loops=1)	
7	[...] -> Hash (cost=152.75..152.75 rows=4723 width=12) (actual time=1.567..1.568 rows=4551 loops=1)	
8	[...] Buckets: 8192 Batches: 1 Memory Usage: 260kB	
9	[...] -> Index Scan using date_idx on "Student" s (cost=0.29..152.75 rows=4723 width=12) (actual time=0.034..0.911 rows=4551 loops=1)	
10	[...] Index Cond: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))	
11	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..1.69 rows=28 width=20) (actual time=0.011..0.014 rows=27 loops=4551)	
12	[...] Index Cond: (amka = (p.amka)::text)	
13	[...] Heap Fetches: 0	
14	[...] -> Hash (cost=7521.03..7521.03 rows=30 width=12) (actual time=61.285..61.288 rows=32 loops=1)	

15	[...] Buckets: 1024 Batches: 1 Memory Usage: 10kB
16	[...] -> Hash Join (cost=49.71..7521.03 rows=30 width=12) (actual time=0.321..61.226 rows=32 loops=1)
17	[...] Hash Cond: (w.module_no = l.module_no)
18	[...] -> Seq Scan on "Workgroup" w (cost=0.00..7470.40 rows=165 width=8) (actual time=0.043..60.837 rows=209 loops=1)
19	[...] Filter: (grade > '8'::numeric)
20	[...] Rows Removed by Filter: 376063
21	[...] -> Hash (cost=45.40..45.40 rows=345 width=4) (actual time=0.269..0.270 rows=345 loops=1)
22	[...] Buckets: 1024 Batches: 1 Memory Usage: 21kB
23	[...] -> Seq Scan on "LabModule" l (cost=0.00..45.40 rows=345 width=4) (actual time=0.009..0.217 rows=345 loops=1)
24	[...] Filter: (type = 'project'::labmodule_type)
25	[...] Rows Removed by Filter: 1527
26	Planning Time: 1.461 ms
27	Execution Time: 166.120 ms

QUERY PLAN		
	text	
1	Hash Join (cost=7733.70..19499.58 rows=1 width=34) (actual time=70.747..161.307 rows=11 loops=1)	
2	[...] Hash Cond: ((j."wglD" = w."wglD") AND (j.module_no = l.module_no))	
3	[...] -> Nested Loop (cost=212.22..10999.31 rows=130503 width=42) (actual time=1.579..91.521 rows=124772 loops=1)	
4	[...] -> Hash Join (cost=211.79..1713.00 rows=4723 width=58) (actual time=1.553..19.328 rows=4551 loops=1)	
5	[...] Hash Cond: ((p.amka)::text = (s.amka)::text)	
6	[...] -> Seq Scan on "Person" p (cost=0.00..1354.78 rows=55778 width=46) (actual time=0.026..3.654 rows=55778 loops=1)	
7	[...] -> Hash (cost=152.75..152.75 rows=4723 width=12) (actual time=1.448..1.449 rows=4551 loops=1)	
8	[...] Buckets: 8192 Batches: 1 Memory Usage: 260kB	
9	[...] -> Index Scan using date_idx on "Student" s (cost=0.29..152.75 rows=4723 width=12) (actual time=0.023..0.765 rows=4551 loops=1)	
10	[...] Index Cond: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))	
11	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..1.69 rows=28 width=20) (actual time=0.011..0.014 rows=27 loops=4551)	
12	[...] Index Cond: (amka = (p.amka)::text)	
13	[...] Heap Fetches: 0	
14	[...] -> Hash (cost=7521.03..7521.03 rows=30 width=12) (actual time=61.138..61.141 rows=32 loops=1)	

15	[...] Buckets: 1024 Batches: 1 Memory Usage: 10kB
16	[...] -> Hash Join (cost=49.71..7521.03 rows=30 width=12) (actual time=0.299..61.104 rows=32 loops=1)
17	[...] Hash Cond: (w.module_no = l.module_no)
18	[...] -> Seq Scan on "Workgroup" w (cost=0.00..7470.40 rows=165 width=8) (actual time=0.037..60.762 rows=209 loops=1)
19	[...] Filter: (grade > '8'::numeric)
20	[...] Rows Removed by Filter: 376063
21	[...] -> Hash (cost=45.40..45.40 rows=345 width=4) (actual time=0.255..0.256 rows=345 loops=1)
22	[...] Buckets: 1024 Batches: 1 Memory Usage: 21kB
23	[...] -> Seq Scan on "LabModule" l (cost=0.00..45.40 rows=345 width=4) (actual time=0.008..0.213 rows=345 loops=1)
24	[...] Filter: (type = 'project'::labmodule_type)
25	[...] Rows Removed by Filter: 1527
26	Planning Time: 1.474 ms
27	Execution Time: 161.504 ms

ΜΕ ΕΥΡΕΤΗΡΙΟ ΤΥΠΟΥ HASH (type)	
Εκτελέσεις	Χρόνος Απόκρισης (ms)
1 ^η	171.190
2 ^η	168.188
3 ^η	172.271
ΜΟ	170.549

Συμπεραίνουμε ότι με τη χρήση του ευρετηρίου Type_idx υπάρχει ελάχιστη έως μηδαμινή μείωση του μέσου χρόνου εκτέλεσης περίπου κατά 1%, γεγονός που οφείλεται στο μικρό selectivity του τύπου εργασίας αλλά και στον σχετικά μικρό αριθμό tuples της βάσης.

QUERY PLAN		🔒
	text	
1	Hash Join (cost=8780.24..20143.38 rows=1 width=34) (actual time=74.665..171.011 rows=11 loops=1)	
2	[...] Hash Cond: ((j.wgID = w.wgID) AND (j.module_no = l.module_no))	
3	[...] -> Nested Loop (cost=1262.42..11696.93 rows=123816 width=42) (actual time=6.201..101.410 rows=124772 loops=1)	
4	[...] -> Hash Join (cost=1261.99..2763.20 rows=4481 width=58) (actual time=6.124..25.875 rows=4551 loops=1)	
5	[...] Hash Cond: ((p.amka)::text = (s.amka)::text)	
6	[...] -> Seq Scan on "Person" p (cost=0.00..1354.78 rows=55778 width=46) (actual time=0.011..3.677 rows=55778 loops=1)	
7	[...] -> Hash (cost=1205.98..1205.98 rows=4481 width=12) (actual time=6.055..6.057 rows=4551 loops=1)	
8	[...] Buckets: 8192 Batches: 1 Memory Usage: 260kB	
9	[...] -> Seq Scan on "Student" s (cost=0.00..1205.98 rows=4481 width=12) (actual time=0.025..5.430 rows=4551 loops=1)	
10	[...] Filter: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))	
11	[...] Rows Removed by Filter: 49381	
12	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..1.70 rows=29 width=20) (actual time=0.011..0.014 rows=27 loops=4551)	
13	[...] Index Cond: (amka = (p.amka)::text)	
14	[...] Heap Fetches: 0	
15	[...] -> Hash (cost=7517.12..7517.12 rows=47 width=12) (actual time=60.887..60.891 rows=32 loops=1)	
16	[...] Buckets: 1024 Batches: 1 Memory Usage: 10kB	
17	[...] -> Hash Join (cost=45.30..7517.12 rows=47 width=12) (actual time=0.296..60.848 rows=32 loops=1)	
18	[...] Hash Cond: (w.module_no = l.module_no)	
19	[...] -> Seq Scan on "Workgroup" w (cost=0.00..7470.40 rows=253 width=8) (actual time=0.061..60.524 rows=209 loops=1)	
20	[...] Filter: (grade > '8'::numeric)	
21	[...] Rows Removed by Filter: 376063	
22	[...] -> Hash (cost=40.99..40.99 rows=345 width=4) (actual time=0.222..0.224 rows=345 loops=1)	
23	[...] Buckets: 1024 Batches: 1 Memory Usage: 21kB	
24	[...] -> Bitmap Heap Scan on "LabModule" l (cost=14.67..40.99 rows=345 width=4) (actual time=0.043..0.180 rows=345 loops=1)	
25	[...] Recheck Cond: (type = 'project'::labmodule_type)	
26	[...] Heap Blocks: exact=22	
27	[...] -> Bitmap Index Scan on type_idx (cost=0.00..14.59 rows=345 width=0) (actual time=0.029..0.029 rows=345 loops=1)	
28	[...] Index Cond: (type = 'project'::labmodule_type)	
29	Planning Time: 1.411 ms	
30	Execution Time: 171.190 ms	

	QUERY PLAN	
	text	
1	Hash Join (cost=8780.24..20143.38 rows=1 width=34) (actual time=76.231..167.995 rows=11 loops=1)	
2	[...] Hash Cond: ((j."wglD" = w."wglD") AND (j.module_no = l.module_no))	
3	[...] -> Nested Loop (cost=1262.42..11696.93 rows=123816 width=42) (actual time=6.116..97.570 rows=124772 loops=1)	
4	[...] -> Hash Join (cost=1261.99..2763.20 rows=4481 width=58) (actual time=6.025..23.433 rows=4551 loops=1)	
5	[...] Hash Cond: ((p.amka)::text = (s.amka)::text)	
6	[...] -> Seq Scan on "Person" p (cost=0.00..1354.78 rows=55778 width=46) (actual time=0.013..3.656 rows=55778 loops=1)	
7	[...] -> Hash (cost=1205.98..1205.98 rows=4481 width=12) (actual time=5.954..5.955 rows=4551 loops=1)	
8	[...] Buckets: 8192 Batches: 1 Memory Usage: 260kB	
9	[...] -> Seq Scan on "Student" s (cost=0.00..1205.98 rows=4481 width=12) (actual time=0.026..5.339 rows=4551 loops=1)	
10	[...] Filter: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))	
11	[...] Rows Removed by Filter: 49381	
12	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..1.70 rows=29 width=20) (actual time=0.011..0.014 rows=27 loops=4551)	
13	[...] Index Cond: (amka = (p.amka)::text)	
14	[...] Heap Fetches: 0	
15	[...] -> Hash (cost=7517.12..7517.12 rows=47 width=12) (actual time=61.766..61.769 rows=32 loops=1)	
16	[...] Buckets: 1024 Batches: 1 Memory Usage: 10kB	
17	[...] -> Hash Join (cost=45.30..7517.12 rows=47 width=12) (actual time=0.351..61.717 rows=32 loops=1)	
18	[...] Hash Cond: (w.module_no = l.module_no)	
19	[...] -> Seq Scan on "Workgroup" w (cost=0.00..7470.40 rows=253 width=8) (actual time=0.070..61.319 rows=209 loops=1)	
20	[...] Filter: (grade > '8'::numeric)	
21	[...] Rows Removed by Filter: 376063	
22	[...] -> Hash (cost=40.99..40.99 rows=345 width=4) (actual time=0.264..0.266 rows=345 loops=1)	
23	[...] Buckets: 1024 Batches: 1 Memory Usage: 21kB	
24	[...] -> Bitmap Heap Scan on "LabModule" l (cost=14.67..40.99 rows=345 width=4) (actual time=0.061..0.220 rows=345 loops=1)	
25	[...] Recheck Cond: (type = 'project'::labmodule_type)	
26	[...] Heap Blocks: exact=22	
27	[...] -> Bitmap Index Scan on type_idx (cost=0.00..14.59 rows=345 width=0) (actual time=0.042..0.043 rows=345 loops=1)	
28	[...] Index Cond: (type = 'project'::labmodule_type)	
29	Planning Time: 1.510 ms	
30	Execution Time: 168.188 ms	

	QUERY PLAN	
	text	
1	Hash Join (cost=8780.24..20143.38 rows=1 width=34) (actual time=74.086..172.114 rows=11 loops=1)	
2	[...] Hash Cond: ((j."wglD" = w."wglD") AND (j.module_no = l.module_no))	
3	[...] -> Nested Loop (cost=1262.42..11696.93 rows=123816 width=42) (actual time=6.082..103.205 rows=124772 loops=1)	
4	[...] -> Hash Join (cost=1261.99..2763.20 rows=4481 width=58) (actual time=6.016..26.513 rows=4551 loops=1)	
5	[...] Hash Cond: ((p.amka)::text = (s.amka)::text)	
6	[...] -> Seq Scan on "Person" p (cost=0.00..1354.78 rows=55778 width=46) (actual time=0.010..3.708 rows=55778 loops=1)	
7	[...] -> Hash (cost=1205.98..1205.98 rows=4481 width=12) (actual time=5.950..5.951 rows=4551 loops=1)	
8	[...] Buckets: 8192 Batches: 1 Memory Usage: 260kB	
9	[...] -> Seq Scan on "Student" s (cost=0.00..1205.98 rows=4481 width=12) (actual time=0.023..5.343 rows=4551 loops=1)	
10	[...] Filter: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))	
11	[...] Rows Removed by Filter: 49381	
12	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..1.70 rows=29 width=20) (actual time=0.011..0.014 rows=27 loops=4551)	
13	[...] Index Cond: (amka = (p.amka)::text)	
14	[...] Heap Fetches: 0	

15	[...] -> Hash (cost=7517.12..7517.12 rows=47 width=12) (actual time=60.150..60.155 rows=32 loops=1)
16	[...] Buckets: 1024 Batches: 1 Memory Usage: 10kB
17	[...] -> Hash Join (cost=45.30..7517.12 rows=47 width=12) (actual time=0.292..60.107 rows=32 loops=1)
18	[...] Hash Cond: (w.module_no = l.module_no)
19	[...] -> Seq Scan on "Workgroup" w (cost=0.00..7470.40 rows=253 width=8) (actual time=0.060..59.770 rows=209 loops=1)
20	[...] Filter: (grade > '8'::numeric)
21	[...] Rows Removed by Filter: 376063
22	[...] -> Hash (cost=40.99..40.99 rows=345 width=4) (actual time=0.221..0.223 rows=345 loops=1)
23	[...] Buckets: 1024 Batches: 1 Memory Usage: 21kB
24	[...] -> Bitmap Heap Scan on "LabModule" l (cost=14.67..40.99 rows=345 width=4) (actual time=0.045..0.179 rows=345 loops=1)
25	[...] Recheck Cond: (type = 'project'::labmodule_type)
26	[...] Heap Blocks: exact=22
27	[...] -> Bitmap Index Scan on type_idx (cost=0.00..14.59 rows=345 width=0) (actual time=0.031..0.031 rows=345 loops=1)
28	[...] Index Cond: (type = 'project'::labmodule_type)
29	Planning Time: 1.274 ms
30	Execution Time: 172.271 ms

ΜΕ ΣΥΝΔΥΑΣΜΟ ΤΩΝ 3 ΕΥΡΕΤΗΡΙΩΝ	
Εκτελέσεις	Χρόνος Απόκρισης (ms)
1 ^η	108.353
2 ^η	94.358
3 ^η	113.533
ΜΟ	105.415

Παρατηρούμε την μεγαλύτερη μείωση στον μέσο χρόνο εκτέλεσης, και συγκεκριμένα 39%. Αυτό είναι αναμενόμενο, καθώς αν αθροίσουμε τα προηγούμενα ποσοστά μείωσης το αποτέλεσμα είναι 34%+4%+1%=39%.

	QUERY PLAN	
	text	
1	Hash Join (cost=272.70..12029.88 rows=1 width=34) (actual time=8.760..108.240 rows=11 loops=1)	
2	[...] Hash Cond: ((j."wgID" = w."wgID") AND (j.module_no = l.module_no))	
3	[...] -> Nested Loop (cost=212.09..10991.32 rows=130392 width=42) (actual time=1.265..99.265 rows=124772 loops=1)	
4	[...] -> Hash Join (cost=211.66..1712.87 rows=4719 width=58) (actual time=1.227..22.774 rows=4551 loops=1)	
5	[...] Hash Cond: ((p.amka)::text = (s.amka)::text)	
6	[...] -> Seq Scan on "Person" p (cost=0.00..1354.78 rows=55778 width=46) (actual time=0.010..3.784 rows=55778 loops=1)	
7	[...] -> Hash (cost=152.67..152.67 rows=4719 width=12) (actual time=1.202..1.204 rows=4551 loops=1)	
8	[...] Buckets: 8192 Batches: 1 Memory Usage: 260kB	
9	[...] -> Index Scan using date_idx on "Student" s (cost=0.29..152.67 rows=4719 width=12) (actual time=0.013..0.691 rows=4551 loops=1)	
10	[...] Index Cond: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))	
11	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..1.69 rows=28 width=20) (actual time=0.012..0.014 rows=27 loops=4551)	
12	[...] Index Cond: (amka = (p.amka)::text)	
13	[...] Heap Fetches: 0	
14	[...] -> Hash (cost=59.99..59.99 rows=42 width=12) (actual time=0.262..0.266 rows=32 loops=1)	

15	[...] Buckets: 1024 Batches: 1 Memory Usage: 10kB
16	[...] -> Hash Join (cost=45.72..59.99 rows=42 width=12) (actual time=0.202..0.259 rows=32 loops=1)
17	[...] Hash Cond: (w.module_no = l.module_no)
18	[...] -> Index Scan using grade_idx on "Workgroup" w (cost=0.42..13.41 rows=228 width=8) (actual time=0.026..0.056 rows=209 loops=1)
19	[...] Index Cond: (grade > '8'::numeric)
20	[...] -> Hash (cost=40.99..40.99 rows=345 width=4) (actual time=0.171..0.173 rows=345 loops=1)
21	[...] Buckets: 1024 Batches: 1 Memory Usage: 21kB
22	[...] -> Bitmap Heap Scan on "LabModule" l (cost=14.67..40.99 rows=345 width=4) (actual time=0.030..0.137 rows=345 loops=1)
23	[...] Recheck Cond: (type = 'project'::labmodule_type)
24	[...] Heap Blocks: exact=22
25	[...] -> Bitmap Index Scan on type_idx (cost=0.00..14.59 rows=345 width=0) (actual time=0.019..0.019 rows=345 loops=1)
26	[...] Index Cond: (type = 'project'::labmodule_type)
27	Planning Time: 1.208 ms
28	Execution Time: 108.353 ms

QUERY PLAN

text

1	Hash Join (cost=272.70..12029.88 rows=1 width=34) (actual time=8.201..94.188 rows=11 loops=1)
2	[...] Hash Cond: ((j."wglD" = w."wglD") AND (j.module_no = l.module_no))
3	[...] -> Nested Loop (cost=212.09..10991.32 rows=130392 width=42) (actual time=1.347..85.381 rows=124772 loops=1)
4	[...] -> Hash Join (cost=211.66..1712.87 rows=4719 width=58) (actual time=1.316..15.449 rows=4551 loops=1)
5	[...] Hash Cond: ((p.amka)::text = (s.amka)::text)
6	[...] -> Seq Scan on "Person" p (cost=0.00..1354.78 rows=55778 width=46) (actual time=0.007..3.547 rows=55778 loops=1)
7	[...] -> Hash (cost=152.67..152.67 rows=4719 width=12) (actual time=1.276..1.276 rows=4551 loops=1)
8	[...] Buckets: 8192 Batches: 1 Memory Usage: 260kB
9	[...] -> Index Scan using date_idx on "Student" s (cost=0.29..152.67 rows=4719 width=12) (actual time=0.011..0.685 rows=4551 loops=1)
10	[...] Index Cond: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))
11	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..1.69 rows=28 width=20) (actual time=0.010..0.013 rows=27 loops=4551)
12	[...] Index Cond: (amka = (p.amka)::text)
13	[...] Heap Fetches: 0
14	[...] -> Hash (cost=59.99..59.99 rows=42 width=12) (actual time=0.273..0.277 rows=32 loops=1)
15	[...] Buckets: 1024 Batches: 1 Memory Usage: 10kB
16	[...] -> Hash Join (cost=45.72..59.99 rows=42 width=12) (actual time=0.213..0.269 rows=32 loops=1)
17	[...] Hash Cond: (w.module_no = l.module_no)
18	[...] -> Index Scan using grade_idx on "Workgroup" w (cost=0.42..13.41 rows=228 width=8) (actual time=0.024..0.051 rows=209 loops=1)
19	[...] Index Cond: (grade > '8'::numeric)
20	[...] -> Hash (cost=40.99..40.99 rows=345 width=4) (actual time=0.185..0.186 rows=345 loops=1)
21	[...] Buckets: 1024 Batches: 1 Memory Usage: 21kB
22	[...] -> Bitmap Heap Scan on "LabModule" l (cost=14.67..40.99 rows=345 width=4) (actual time=0.034..0.140 rows=345 loops=1)
23	[...] Recheck Cond: (type = 'project'::labmodule_type)
24	[...] Heap Blocks: exact=22
25	[...] -> Bitmap Index Scan on type_idx (cost=0.00..14.59 rows=345 width=0) (actual time=0.023..0.023 rows=345 loops=1)
26	[...] Index Cond: (type = 'project'::labmodule_type)
27	Planning Time: 1.507 ms
28	Execution Time: 94.358 ms

	QUERY PLAN text	
1	Hash Join (cost=272.70..12029.88 rows=1 width=34) (actual time=9.410..113.422 rows=11 loops=1)	
2	[...] Hash Cond: ((j."wglD" = w."wglD") AND (j.module_no = l.module_no))	
3	[...] -> Nested Loop (cost=212.09..10991.32 rows=130392 width=42) (actual time=1.319..103.959 rows=124772 loops=1)	
4	[...] -> Hash Join (cost=211.66..1712.87 rows=4719 width=58) (actual time=1.284..23.980 rows=4551 loops=1)	
5	[...] Hash Cond: ((p.amka)::text = (s.amka)::text)	
6	[...] -> Seq Scan on "Person" p (cost=0.00..1354.78 rows=55778 width=46) (actual time=0.013..3.800 rows=55778 loops=1)	
7	[...] -> Hash (cost=152.67..152.67 rows=4719 width=12) (actual time=1.252..1.253 rows=4551 loops=1)	
8	[...] Buckets: 8192 Batches: 1 Memory Usage: 260kB	
9	[...] -> Index Scan using date_idx on "Student" s (cost=0.29..152.67 rows=4719 width=12) (actual time=0.014..0.711 rows=4551 loops=1)	
10	[...] Index Cond: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))	
11	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..1.69 rows=28 width=20) (actual time=0.012..0.015 rows=27 loops=4551)	
12	[...] Index Cond: (amka = (p.amka)::text)	
13	[...] Heap Fetches: 0	
14	[...] -> Hash (cost=59.99..59.99 rows=42 width=12) (actual time=0.405..0.409 rows=32 loops=1)	
15	[...] Buckets: 1024 Batches: 1 Memory Usage: 10kB	
16	[...] -> Hash Join (cost=45.72..59.99 rows=42 width=12) (actual time=0.304..0.400 rows=32 loops=1)	
17	[...] Hash Cond: (w.module_no = l.module_no)	
18	[...] -> Index Scan using grade_idx on "Workgroup" w (cost=0.42..13.41 rows=228 width=8) (actual time=0.037..0.086 rows=209 loops=1)	
19	[...] Index Cond: (grade > '8'::numeric)	
20	[...] -> Hash (cost=40.99..40.99 rows=345 width=4) (actual time=0.260..0.262 rows=345 loops=1)	
21	[...] Buckets: 1024 Batches: 1 Memory Usage: 21kB	
22	[...] -> Bitmap Heap Scan on "LabModule" l (cost=14.67..40.99 rows=345 width=4) (actual time=0.039..0.201 rows=345 loops=1)	
23	[...] Recheck Cond: (type = 'project'::labmodule_type)	
24	[...] Heap Blocks: exact=22	
25	[...] -> Bitmap Index Scan on type_idx (cost=0.00..14.59 rows=345 width=0) (actual time=0.025..0.025 rows=345 loops=1)	
26	[...] Index Cond: (type = 'project'::labmodule_type)	
27	Planning Time: 1.868 ms	
28	Execution Time: 113.553 ms	

Έπειτα, για καθένα από τα ευρετήρια τύπου btree αξιοποιήσαμε την δυνατότητα ομαδοποίησης (clustering).

ΜΕ CLUSTER ΣΤΟ ΕΥΡΕΤΗΡΙΟ BTREE (grade)	
Εκτελέσεις	Χρόνος Απόκρισης (ms)
1 ^η	81.072
2 ^η	81.713
3 ^η	79.876
ΜΟ	80.887

Παρατηρήσαμε αξιοσημείωτη βελτίωση του χρόνου εκτέλεσης, καθώς με την συσταδοποίηση του Grade_idx ο χρόνος μειώθηκε περίπου κατά 29%, γεγονός αναμενόμενο από τη θεωρία.

	QUERY PLAN	
	text	
1	Hash Join (cost=1328.69..12940.26 rows=1 width=34) (actual time=10.605..80.975 rows=11 loops=1)	
2	[...] Hash Cond: ((j."wglD" = w."wglD") AND (j.module_no = l.module_no))	
3	[...] -> Nested Loop (cost=1264.56..11912.06 rows=128541 width=42) (actual time=4.841..73.445 rows=124772 loops=1)	
4	[...] -> Hash Join (cost=1264.13..2765.34 rows=4652 width=58) (actual time=4.799..15.809 rows=4551 loops=1)	
5	[...] Hash Cond: ((p.amka)::text = (s.amka)::text)	
6	[...] -> Seq Scan on "Person" p (cost=0.00..1354.78 rows=55778 width=46) (actual time=0.005..2.961 rows=55778 loops=1)	
7	[...] -> Hash (cost=1205.98..1205.98 rows=4652 width=12) (actual time=4.769..4.770 rows=4551 loops=1)	
8	[...] Buckets: 8192 Batches: 1 Memory Usage: 260kB	
9	[...] -> Seq Scan on "Student" s (cost=0.00..1205.98 rows=4652 width=12) (actual time=0.017..4.286 rows=4551 loops=1)	
10	[...] Filter: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))	
11	[...] Rows Removed by Filter: 49381	
12	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..1.69 rows=28 width=20) (actual time=0.008..0.011 rows=27 loops=4551)	
13	[...] Index Cond: (amka = (p.amka)::text)	
14	[...] Heap Fetches: 0	
15	[...] -> Hash (cost=63.60..63.60 rows=36 width=12) (actual time=0.291..0.293 rows=32 loops=1)	
16	[...] Buckets: 1024 Batches: 1 Memory Usage: 10kB	
17	[...] -> Hash Join (cost=50.14..63.60 rows=36 width=12) (actual time=0.242..0.288 rows=32 loops=1)	
18	[...] Hash Cond: (w.module_no = l.module_no)	
19	[...] -> Index Scan using grade_idx on "Workgroup" w (cost=0.42..12.80 rows=193 width=8) (actual time=0.026..0.048 rows=209 loops=1)	
20	[...] Index Cond: (grade > '8'::numeric)	
21	[...] -> Hash (cost=45.40..45.40 rows=345 width=4) (actual time=0.210..0.211 rows=345 loops=1)	
22	[...] Buckets: 1024 Batches: 1 Memory Usage: 21kB	
23	[...] -> Seq Scan on "LabModule" l (cost=0.00..45.40 rows=345 width=4) (actual time=0.014..0.176 rows=345 loops=1)	
24	[...] Filter: (type = 'project'::labmodule_type)	
25	[...] Rows Removed by Filter: 1527	
26	Planning Time: 1.144 ms	
27	Execution Time: 81.072 ms	

	QUERY PLAN	
	text	
1	Hash Join (cost=1328.69..12940.26 rows=1 width=34) (actual time=11.275..81.604 rows=11 loops=1)	
2	[...] Hash Cond: ((j."wglD" = w."wglD") AND (j.module_no = l.module_no))	
3	[...] -> Nested Loop (cost=1264.56..11912.06 rows=128541 width=42) (actual time=5.227..74.049 rows=124772 loops=1)	
4	[...] -> Hash Join (cost=1264.13..2765.34 rows=4652 width=58) (actual time=5.182..16.612 rows=4551 loops=1)	
5	[...] Hash Cond: ((p.amka)::text = (s.amka)::text)	
6	[...] -> Seq Scan on "Person" p (cost=0.00..1354.78 rows=55778 width=46) (actual time=0.007..2.953 rows=55778 loops=1)	
7	[...] -> Hash (cost=1205.98..1205.98 rows=4652 width=12) (actual time=5.143..5.143 rows=4551 loops=1)	
8	[...] Buckets: 8192 Batches: 1 Memory Usage: 260kB	
9	[...] -> Seq Scan on "Student" s (cost=0.00..1205.98 rows=4652 width=12) (actual time=0.022..4.591 rows=4551 loops=1)	
10	[...] Filter: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))	
11	[...] Rows Removed by Filter: 49381	
12	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..1.69 rows=28 width=20) (actual time=0.008..0.011 rows=27 loops=4551)	
13	[...] Index Cond: (amka = (p.amka)::text)	
14	[...] Heap Fetches: 0	

15	[...] -> Hash (cost=63.60..63.60 rows=36 width=12) (actual time=0.346..0.348 rows=32 loops=1)
16	[...] Buckets: 1024 Batches: 1 Memory Usage: 10kB
17	[...] -> Hash Join (cost=50.14..63.60 rows=36 width=12) (actual time=0.287..0.341 rows=32 loops=1)
18	[...] Hash Cond: (w.module_no = l.module_no)
19	[...] -> Index Scan using grade_idx on "Workgroup" w (cost=0.42..12.80 rows=193 width=8) (actual time=0.031..0.059 rows=209 loops=1)
20	[...] Index Cond: (grade > '8'::numeric)
21	[...] -> Hash (cost=45.40..45.40 rows=345 width=4) (actual time=0.249..0.249 rows=345 loops=1)
22	[...] Buckets: 1024 Batches: 1 Memory Usage: 21kB
23	[...] -> Seq Scan on "LabModule" l (cost=0.00..45.40 rows=345 width=4) (actual time=0.021..0.208 rows=345 loops=1)
24	[...] Filter: (type = 'project'::labmodule_type)
25	[...] Rows Removed by Filter: 1527
26	Planning Time: 1.261 ms
27	Execution Time: 81.713 ms

QUERY PLAN

text



1	Hash Join (cost=1328.69..12940.26 rows=1 width=34) (actual time=10.445..79.778 rows=11 loops=1)
2	[...] Hash Cond: ((j."wglD" = w."wglD") AND (j.module_no = l.module_no))
3	[...] -> Nested Loop (cost=1264.56..11912.06 rows=128541 width=42) (actual time=4.690..72.358 rows=124772 loops=1)
4	[...] -> Hash Join (cost=1264.13..2765.34 rows=4652 width=58) (actual time=4.645..15.631 rows=4551 loops=1)
5	[...] Hash Cond: ((p.amka)::text = (s.amka)::text)
6	[...] -> Seq Scan on "Person" p (cost=0.00..1354.78 rows=55778 width=46) (actual time=0.004..2.960 rows=55778 loops=1)
7	[...] -> Hash (cost=1205.98..1205.98 rows=4652 width=12) (actual time=4.614..4.614 rows=4551 loops=1)
8	[...] Buckets: 8192 Batches: 1 Memory Usage: 260kB
9	[...] -> Seq Scan on "Student" s (cost=0.00..1205.98 rows=4652 width=12) (actual time=0.017..4.155 rows=4551 loops=1)
10	[...] Filter: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))
11	[...] Rows Removed by Filter: 49381
12	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..1.69 rows=28 width=20) (actual time=0.008..0.010 rows=27 loops=4551)
13	[...] Index Cond: (amka = (p.amka)::text)
14	[...] Heap Fetches: 0

15	[...] -> Hash (cost=63.60..63.60 rows=36 width=12) (actual time=0.281..0.282 rows=32 loops=1)
16	[...] Buckets: 1024 Batches: 1 Memory Usage: 10kB
17	[...] -> Hash Join (cost=50.14..63.60 rows=36 width=12) (actual time=0.234..0.277 rows=32 loops=1)
18	[...] Hash Cond: (w.module_no = l.module_no)
19	[...] -> Index Scan using grade_idx on "Workgroup" w (cost=0.42..12.80 rows=193 width=8) (actual time=0.022..0.044 rows=209 loops=1)
20	[...] Index Cond: (grade > '8'::numeric)
21	[...] -> Hash (cost=45.40..45.40 rows=345 width=4) (actual time=0.207..0.207 rows=345 loops=1)
22	[...] Buckets: 1024 Batches: 1 Memory Usage: 21kB
23	[...] -> Seq Scan on "LabModule" l (cost=0.00..45.40 rows=345 width=4) (actual time=0.008..0.172 rows=345 loops=1)
24	[...] Filter: (type = 'project'::labmodule_type)
25	[...] Rows Removed by Filter: 1527
26	Planning Time: 1.102 ms
27	Execution Time: 79.876 ms

ΜΕ CLUSTER ΣΤΟ ΕΥΡΕΤΗΡΙΟ BTREE (entry_date)	
Εκτελέσεις	Χρόνος Απόκρισης (ms)
1 ^η	122.898
2 ^η	127.758
3 ^η	125.270
ΜΟ	125.309

Κι εδώ παρατηρούμε βελτίωση του χρόνου εκτέλεσης, καθώς με την συσταδοποίηση του Type_idx ο χρόνος μειώθηκε περίπου κατά 24%, γεγονός αναμενόμενο από τη θεωρία.

QUERY PLAN		🔒
	text	
1	Hash Join (cost=7733.95..19508.52 rows=1 width=34) (actual time=54.482..122.789 rows=11 loops=1)	
2	[...] Hash Cond: ((j."wglD" = w."wglD") AND (j.module_no = l.module_no))	
3	[...] -> Nested Loop (cost=212.35..11007.31 rows=130613 width=42) (actual time=1.344..67.795 rows=124772 loops=1)	
4	[...] -> Hash Join (cost=211.92..1713.13 rows=4727 width=58) (actual time=1.307..12.233 rows=4551 loops=1)	
5	[...] Hash Cond: ((p.amka)::text = (s.amka)::text)	
6	[...] -> Seq Scan on "Person" p (cost=0.00..1354.78 rows=55778 width=46) (actual time=0.023..2.936 rows=55778 loops=1)	
7	[...] -> Hash (cost=152.83..152.83 rows=4727 width=12) (actual time=1.240..1.241 rows=4551 loops=1)	
8	[...] Buckets: 8192 Batches: 1 Memory Usage: 260kB	
9	[...] -> Index Scan using date_idx on "Student" s (cost=0.29..152.83 rows=4727 width=12) (actual time=0.019..0.661 rows=4551 loops=1)	
10	[...] Index Cond: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))	
11	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..1.69 rows=28 width=20) (actual time=0.008..0.010 rows=27 loops=4551)	
12	[...] Index Cond: (amka = (p.amka)::text)	
13	[...] Heap Fetches: 0	
14	[...] -> Hash (cost=7521.11..7521.11 rows=33 width=12) (actual time=47.742..47.744 rows=32 loops=1)	
15	[...] Buckets: 1024 Batches: 1 Memory Usage: 10kB	
16	[...] -> Hash Join (cost=49.71..7521.11 rows=33 width=12) (actual time=47.686..47.736 rows=32 loops=1)	
17	[...] Hash Cond: (w.module_no = l.module_no)	
18	[...] -> Seq Scan on "Workgroup" w (cost=0.00..7470.40 rows=179 width=8) (actual time=47.448..47.483 rows=209 loops=1)	
19	[...] Filter: (grade > '8'::numeric)	
20	[...] Rows Removed by Filter: 376063	
21	[...] -> Hash (cost=45.40..45.40 rows=345 width=4) (actual time=0.219..0.220 rows=345 loops=1)	
22	[...] Buckets: 1024 Batches: 1 Memory Usage: 21kB	
23	[...] -> Seq Scan on "LabModule" l (cost=0.00..45.40 rows=345 width=4) (actual time=0.021..0.180 rows=345 loops=1)	
24	[...] Filter: (type = 'project'::labmodule_type)	
25	[...] Rows Removed by Filter: 1527	
26	Planning Time: 1.281 ms	
27	Execution Time: 122.898 ms	

	QUERY PLAN text
1	Hash Join (cost=7733.95..19508.52 rows=1 width=34) (actual time=54.989..127.651 rows=11 loops=1)
2	[...] Hash Cond: ((j."wglD" = w."wglD") AND (j.module_no = l.module_no))
3	[...] -> Nested Loop (cost=212.35..11007.31 rows=130613 width=42) (actual time=1.444..72.069 rows=124772 loops=1)
4	[...] -> Hash Join (cost=211.92..1713.13 rows=4727 width=58) (actual time=1.415..13.521 rows=4551 loops=1)
5	[...] Hash Cond: ((p.amka)::text = (s.amka)::text)
6	[...] -> Seq Scan on "Person" p (cost=0.00..1354.78 rows=55778 width=46) (actual time=0.026..3.057 rows=55778 loops=1)
7	[...] -> Hash (cost=152.83..152.83 rows=4727 width=12) (actual time=1.347..1.348 rows=4551 loops=1)
8	[...] Buckets: 8192 Batches: 1 Memory Usage: 260kB
9	[...] -> Index Scan using date_idx on "Student" s (cost=0.29..152.83 rows=4727 width=12) (actual time=0.017..0.711 rows=4551 loops=1)
10	[...] Index Cond: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))
11	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..1.69 rows=28 width=20) (actual time=0.008..0.011 rows=27 loops=4551)
12	[...] Index Cond: (amka = (p.amka)::text)
13	[...] Heap Fetches: 0
14	[...] -> Hash (cost=7521.11..7521.11 rows=33 width=12) (actual time=48.087..48.089 rows=32 loops=1)
15	[...] Buckets: 1024 Batches: 1 Memory Usage: 10kB
16	[...] -> Hash Join (cost=49.71..7521.11 rows=33 width=12) (actual time=48.020..48.069 rows=32 loops=1)
17	[...] Hash Cond: (w.module_no = l.module_no)
18	[...] -> Seq Scan on "Workgroup" w (cost=0.00..7470.40 rows=179 width=8) (actual time=47.776..47.807 rows=209 loops=1)
19	[...] Filter: (grade > '8'::numeric)
20	[...] Rows Removed by Filter: 376063
21	[...] -> Hash (cost=45.40..45.40 rows=345 width=4) (actual time=0.224..0.225 rows=345 loops=1)
22	[...] Buckets: 1024 Batches: 1 Memory Usage: 21kB
23	[...] -> Seq Scan on "LabModule" l (cost=0.00..45.40 rows=345 width=4) (actual time=0.029..0.194 rows=345 loops=1)
24	[...] Filter: (type = 'project'::labmodule_type)
25	[...] Rows Removed by Filter: 1527
26	Planning Time: 1.489 ms
27	Execution Time: 127.758 ms

	QUERY PLAN text
1	Hash Join (cost=7733.95..19508.52 rows=1 width=34) (actual time=55.683..125.131 rows=11 loops=1)
2	[...] Hash Cond: ((j."wglD" = w."wglD") AND (j.module_no = l.module_no))
3	[...] -> Nested Loop (cost=212.35..11007.31 rows=130613 width=42) (actual time=1.684..69.513 rows=124772 loops=1)
4	[...] -> Hash Join (cost=211.92..1713.13 rows=4727 width=58) (actual time=1.657..13.006 rows=4551 loops=1)
5	[...] Hash Cond: ((p.amka)::text = (s.amka)::text)
6	[...] -> Seq Scan on "Person" p (cost=0.00..1354.78 rows=55778 width=46) (actual time=0.025..2.960 rows=55778 loops=1)
7	[...] -> Hash (cost=152.83..152.83 rows=4727 width=12) (actual time=1.582..1.583 rows=4551 loops=1)
8	[...] Buckets: 8192 Batches: 1 Memory Usage: 260kB
9	[...] -> Index Scan using date_idx on "Student" s (cost=0.29..152.83 rows=4727 width=12) (actual time=0.023..0.829 rows=4551 loops=1)
10	[...] Index Cond: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))
11	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..1.69 rows=28 width=20) (actual time=0.008..0.010 rows=27 loops=4551)
12	[...] Index Cond: (amka = (p.amka)::text)
13	[...] Heap Fetches: 0
14	[...] -> Hash (cost=7521.11..7521.11 rows=33 width=12) (actual time=48.480..48.482 rows=32 loops=1)

15	[...] Buckets: 1024 Batches: 1 Memory Usage: 10kB
16	[...] -> Hash Join (cost=49.71..7521.11 rows=33 width=12) (actual time=48.419..48.471 rows=32 loops=1)
17	[...] Hash Cond: (w.module_no = l.module_no)
18	[...] -> Seq Scan on "Workgroup" w (cost=0.00..7470.40 rows=179 width=8) (actual time=48.171..48.205 rows=209 loops=1)
19	[...] Filter: (grade > '8'::numeric)
20	[...] Rows Removed by Filter: 376063
21	[...] -> Hash (cost=45.40..45.40 rows=345 width=4) (actual time=0.229..0.229 rows=345 loops=1)
22	[...] Buckets: 1024 Batches: 1 Memory Usage: 21kB
23	[...] -> Seq Scan on "LabModule" l (cost=0.00..45.40 rows=345 width=4) (actual time=0.020..0.188 rows=345 loops=1)
24	[...] Filter: (type = 'project'::labmodule_type)
25	[...] Rows Removed by Filter: 1527
26	Planning Time: 2.044 ms
27	Execution Time: 125.270 ms

Στη συνέχεια, έγινε απενεργοποίηση κάποιων αλγορίθμων υπολογισμού συνδέσεων τόσο πριν, όσο και μετά την δημιουργία των ευρετηρίων.

ΧΩΡΙΣ ΕΥΡΕΤΗΡΙΑ, ΧΩΡΙΣ hashjoin	
Εκτελέσεις	Χρόνος Απόκρισης (ms)
1 ^η	380.271
2 ^η	377.606
3 ^η	396.022
ΜΟ	384.633

Η αύξηση του χρόνου εκτέλεσης είναι εμφανής και συγκεκριμένα αυξήθηκε κατά 123%, όπως και αναμέναμε εφόσον η χρήση του hashjoin βελτιώνει σημαντικά την απόδοση της εκτέλεσης.

QUERY PLAN		
text		
1	Nested Loop (cost=38490.25..39485.27 rows=1 width=34) (actual time=335.972..377.978 rows=11 loops=1)	
2	[...] Join Filter: (w.module_no = l.module_no)	
3	[...] -> Merge Join (cost=38489.98..39470.93 rows=1 width=42) (actual time=332.226..373.246 rows=76 loops=1)	
4	[...] Merge Cond: ((j.module_no = w.module_no) AND (j."wgID" = w."wgID"))	
5	[...] -> Sort (cost=31012.88..31339.41 rows=130613 width=42) (actual time=272.865..305.355 rows=124772 loops=1)	
6	[...] Sort Key: j.module_no, j."wgID"	
7	[...] Sort Method: external merge Disk: 6584kB	
8	[...] -> Nested Loop (cost=1495.42..15896.08 rows=130613 width=42) (actual time=21.531..154.694 rows=124772 loops=1)	
9	[...] -> Merge Join (cost=1494.99..6601.90 rows=4727 width=58) (actual time=21.475..88.929 rows=4551 loops=1)	
10	[...] Merge Cond: ((p.amka)::text = (s.amka)::text)	
11	[...] -> Index Scan using "Person_pkey" on "Person" p (cost=0.41..4897.06 rows=55778 width=46) (actual time=0.015..41.110 rows=55623 loops=1)	
12	[...] -> Sort (cost=1494.49..1506.30 rows=4727 width=12) (actual time=21.423..22.618 rows=4551 loops=1)	
13	[...] Sort Key: s.amka	
14	[...] Sort Method: quicksort Memory: 406kB	

15	[...] -> Seq Scan on "Student" s (cost=0.00..1205.98 rows=4727 width=12) (actual time=0.043..5.583 rows=4551 loops=1)
16	[...] Filter: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))
17	[...] Rows Removed by Filter: 49381
18	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..1.69 rows=28 width=20) (actual time=0.009..0.012 rows=27 loops=4551)
19	[...] Index Cond: (amka = (p.amka)::text)
20	[...] Heap Fetches: 0
21	[...] -> Sort (cost=7477.10..7477.55 rows=179 width=8) (actual time=58.291..58.307 rows=219 loops=1)
22	[...] Sort Key: w.module_no, w.wgID
23	[...] Sort Method: quicksort Memory: 34kB
24	[...] -> Seq Scan on "Workgroup" w (cost=0.00..7470.40 rows=179 width=8) (actual time=58.171..58.213 rows=209 loops=1)
25	[...] Filter: (grade > '8'::numeric)
26	[...] Rows Removed by Filter: 376063
27	[...] -> Index Scan using "LabModule_pkey" on "LabModule" l (cost=0.28..14.33 rows=1 width=4) (actual time=0.059..0.062 rows=0 loops=76)
28	[...] Index Cond: (module_no = j.module_no)
29	[...] Filter: (type = 'project'::labmodule_type)
30	[...] Rows Removed by Filter: 1
31	Planning Time: 1.861 ms
32	Execution Time: 380.271 ms

	QUERY PLAN	
	text	
1	Nested Loop (cost=38228.76..39208.40 rows=1 width=34) (actual time=332.620..375.183 rows=11 loops=1)	
2	[...] Join Filter: (w.module_no = l.module_no)	
3	[...] -> Merge Join (cost=38228.49..39194.06 rows=1 width=42) (actual time=328.754..370.637 rows=76 loops=1)	
4	[...] Merge Cond: ((j.module_no = w.module_no) AND (j."wgID" = w."wgID"))	
5	[...] -> Sort (cost=30755.16..31076.79 rows=128652 width=42) (actual time=269.211..302.368 rows=124772 loops=1)	
6	[...] Sort Key: j.module_no, j."wgID"	
7	[...] Sort Method: external merge Disk: 6584kB	
8	[...] -> Nested Loop (cost=1490.49..15878.54 rows=128652 width=42) (actual time=23.238..153.655 rows=124772 loops=1)	
9	[...] -> Merge Join (cost=1490.06..6595.92 rows=4656 width=58) (actual time=23.169..89.207 rows=4551 loops=1)	
10	[...] Merge Cond: ((p.amka)::text = (s.amka)::text)	
11	[...] -> Index Scan using "Person_pkey" on "Person" p (cost=0.41..4896.99 rows=55778 width=46) (actual time=0.017..39.628 rows=55623 loops=1)	
12	[...] -> Sort (cost=1489.64..1501.28 rows=4656 width=12) (actual time=23.111..24.285 rows=4551 loops=1)	
13	[...] Sort Key: s.amka	
14	[...] Sort Method: quicksort Memory: 406kB	

15	[...] -> Seq Scan on "Student" s (cost=0.00..1205.98 rows=4656 width=12) (actual time=0.059..5.628 rows=4551 loops=1)
16	[...] Filter: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))
17	[...] Rows Removed by Filter: 49381
18	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..1.70 rows=29 width=20) (actual time=0.009..0.012 rows=27 loops=4551)
19	[...] Index Cond: (amka = (p.amka)::text)
20	[...] Heap Fetches: 0
21	[...] -> Sort (cost=7473.32..7473.55 rows=90 width=8) (actual time=58.409..58.428 rows=219 loops=1)
22	[...] Sort Key: w.module_no, w.wgID
23	[...] Sort Method: quicksort Memory: 34kB
24	[...] -> Seq Scan on "Workgroup" w (cost=0.00..7470.40 rows=90 width=8) (actual time=58.285..58.330 rows=209 loops=1)
25	[...] Filter: (grade > '8'::numeric)
26	[...] Rows Removed by Filter: 376063
27	[...] -> Index Scan using "LabModule_pkey" on "LabModule" l (cost=0.28..14.33 rows=1 width=4) (actual time=0.056..0.059 rows=0 loops=76)
28	[...] Index Cond: (module_no = j.module_no)
29	[...] Filter: (type = 'project'::labmodule_type)
30	[...] Rows Removed by Filter: 1
31	Planning Time: 1.727 ms
32	Execution Time: 377.606 ms

	QUERY PLAN	
	text	
1	Nested Loop (cost=38228.76..39208.40 rows=1 width=34) (actual time=350.936..392.597 rows=11 loops=1)	
2	[...] Join Filter: (w.module_no = l.module_no)	
3	[...] -> Merge Join (cost=38228.49..39194.06 rows=1 width=42) (actual time=347.170..387.946 rows=76 loops=1)	
4	[...] Merge Cond: ((j.module_no = w.module_no) AND (j."wglD" = w."wglD"))	
5	[...] -> Sort (cost=30755.16..31076.79 rows=128652 width=42) (actual time=287.758..320.174 rows=124772 loops=1)	
6	[...] Sort Key: j.module_no,j."wglD"	
7	[...] Sort Method: external merge Disk: 6584kB	
8	[...] -> Nested Loop (cost=1490.49..15878.54 rows=128652 width=42) (actual time=21.715..165.550 rows=124772 loops=1)	
9	[...] -> Merge Join (cost=1490.06..6595.92 rows=4656 width=58) (actual time=21.673..96.830 rows=4551 loops=1)	
10	[...] Merge Cond: ((p.amka)::text = (s.amka)::text)	
11	[...] -> Index Scan using "Person_pkey" on "Person" p (cost=0.41..4896.99 rows=55778 width=46) (actual time=0.011..47.659 rows=55623 loops=1)	
12	[...] -> Sort (cost=1489.64..1501.28 rows=4656 width=12) (actual time=21.630..22.843 rows=4551 loops=1)	
13	[...] Sort Key: s.amka	
14	[...] Sort Method: quicksort Memory: 406kB	
15	[...] -> Seq Scan on "Student" s (cost=0.00..1205.98 rows=4656 width=12) (actual time=0.050..5.429 rows=4551 loops=1)	
16	[...] Filter: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))	
17	[...] Rows Removed by Filter: 49381	
18	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..1.70 rows=29 width=20) (actual time=0.009..0.013 rows=27 loops=4551)	
19	[...] Index Cond: (amka = (p.amka)::text)	
20	[...] Heap Fetches: 0	
21	[...] -> Sort (cost=7473.32..7473.55 rows=90 width=8) (actual time=58.329..58.346 rows=219 loops=1)	
22	[...] Sort Key: w.module_no, w."wglD"	
23	[...] Sort Method: quicksort Memory: 34kB	
24	[...] -> Seq Scan on "Workgroup" w (cost=0.00..7470.40 rows=90 width=8) (actual time=58.221..58.265 rows=209 loops=1)	
25	[...] Filter: (grade > '8'::numeric)	
26	[...] Rows Removed by Filter: 376063	
27	[...] -> Index Scan using "LabModule_pkey" on "LabModule" l (cost=0.28..14.33 rows=1 width=4) (actual time=0.058..0.061 rows=0 loops=76)	
28	[...] Index Cond: (module_no = j.module_no)	
29	[...] Filter: (type = 'project'::labmodule_type)	
30	[...] Rows Removed by Filter: 1	
31	Planning Time: 1.742 ms	
32	Execution Time: 395.022 ms	

ΧΩΡΙΣ ΕΥΡΕΤΗΡΙΑ, ΧΩΡΙΣ hashjoin, ΧΩΡΙΣ mergejoin	
Εκτελέσεις	Χρόνος Απόκρισης (ms)
1 ^η	594.875
2 ^η	608.294
3 ^η	602.232
ΜΟ	601.800

Μετά την απενεργοποίηση του mergejoin η απόδοση πέφτει ακόμα περισσότερο και συγκεκριμένα ο μέσος χρόνος εκτέλεσης αυξάνεται κατά 249% σε σχέση με τον αρχικό. Το αποτέλεσμα αυτό είναι το αναμενόμενο.

	QUERY PLAN text	
1	Nested Loop (cost=0.84..142903.76 rows=1 width=34) (actual time=88.577..594.777 rows=11 loops=1)	
2	[...] Join Filter: ((w."wgID" = j."wgID") AND (l.module_no = j.module_no))	
3	[...] Rows Removed by Join Filter: 3992693	
4	[...] -> Nested Loop (cost=0.84..16086.47 rows=124645 width=42) (actual time=0.134..127.449 rows=124772 loops=1)	
5	[...] -> Nested Loop (cost=0.41..7216.99 rows=4511 width=58) (actual time=0.107..50.691 rows=4551 loops=1)	
6	[...] -> Seq Scan on "Student" s (cost=0.00..1205.98 rows=4511 width=12) (actual time=0.056..6.124 rows=4551 loops=1)	
7	[...] Filter: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))	
8	[...] Rows Removed by Filter: 49381	
9	[...] -> Index Scan using "Person_pkey" on "Person" p (cost=0.41..1.33 rows=1 width=46) (actual time=0.009..0.009 rows=1 loops=4551)	
10	[...] Index Cond: ((amka)::text = (s.amka)::text)	
11	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..1.69 rows=28 width=20) (actual time=0.011..0.014 rows=27 loops=4551)	
12	[...] Index Cond: (amka = (p.amka)::text)	
13	[...] Heap Fetches: 0	
14	[...] -> Materialize (cost=0.00..9027.90 rows=54 width=12) (actual time=0.001..0.002 rows=32 loops=124772)	
15	[...] -> Nested Loop (cost=0.00..9027.63 rows=54 width=12) (actual time=57.815..65.094 rows=32 loops=1)	
16	[...] Join Filter: (w.module_no = l.module_no)	
17	[...] Rows Removed by Join Filter: 72073	
18	[...] -> Seq Scan on "LabModule" l (cost=0.00..45.40 rows=345 width=4) (actual time=0.010..0.231 rows=345 loops=1)	
19	[...] Filter: (type = 'project'::labmodule_type)	
20	[...] Rows Removed by Filter: 1527	
21	[...] -> Materialize (cost=0.00..7471.86 rows=292 width=8) (actual time=0.168..0.175 rows=209 loops=345)	
22	[...] -> Seq Scan on "Workgroup" w (cost=0.00..7470.40 rows=292 width=8) (actual time=57.794..57.838 rows=209 loops=1)	
23	[...] Filter: (grade > '8'::numeric)	
24	[...] Rows Removed by Filter: 376063	
25	Planning Time: 0.690 ms	
26	Execution Time: 594.875 ms	

	QUERY PLAN text	
1	Nested Loop (cost=0.84..142903.76 rows=1 width=34) (actual time=94.298..608.205 rows=11 loops=1)	
2	[...] Join Filter: ((w."wgID" = j."wgID") AND (l.module_no = j.module_no))	
3	[...] Rows Removed by Join Filter: 3992693	
4	[...] -> Nested Loop (cost=0.84..16086.47 rows=124645 width=42) (actual time=0.121..132.684 rows=124772 loops=1)	
5	[...] -> Nested Loop (cost=0.41..7216.99 rows=4511 width=58) (actual time=0.096..52.517 rows=4551 loops=1)	
6	[...] -> Seq Scan on "Student" s (cost=0.00..1205.98 rows=4511 width=12) (actual time=0.045..6.334 rows=4551 loops=1)	
7	[...] Filter: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))	
8	[...] Rows Removed by Filter: 49381	
9	[...] -> Index Scan using "Person_pkey" on "Person" p (cost=0.41..1.33 rows=1 width=46) (actual time=0.010..0.010 rows=1 loops=4551)	
10	[...] Index Cond: ((amka)::text = (s.amka)::text)	
11	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..1.69 rows=28 width=20) (actual time=0.012..0.015 rows=27 loops=4551)	
12	[...] Index Cond: (amka = (p.amka)::text)	
13	[...] Heap Fetches: 0	
14	[...] -> Materialize (cost=0.00..9027.90 rows=54 width=12) (actual time=0.001..0.002 rows=32 loops=124772)	

15	[...] -> Nested Loop (cost=0.00..9027.63 rows=54 width=12) (actual time=62.927..70.624 rows=32 loops=1)
16	[...] Join Filter: (w.module_no = l.module_no)
17	[...] Rows Removed by Join Filter: 72073
18	[...] -> Seq Scan on "LabModule" l (cost=0.00..45.40 rows=345 width=4) (actual time=0.008..0.241 rows=345 loops=1)
19	[...] Filter: (type = 'project'::labmodule_type)
20	[...] Rows Removed by Filter: 1527
21	[...] -> Materialize (cost=0.00..7471.86 rows=292 width=8) (actual time=0.182..0.190 rows=209 loops=345)
22	[...] -> Seq Scan on "Workgroup" w (cost=0.00..7470.40 rows=292 width=8) (actual time=62.909..62.952 rows=209 loops=1)
23	[...] Filter: (grade > '8'::numeric)
24	[...] Rows Removed by Filter: 376063
25	Planning Time: 0.696 ms
26	Execution Time: 608.294 ms

	QUERY PLAN	
	text	
1	Nested Loop (cost=0.84..142903.76 rows=1 width=34) (actual time=87.265..602.159 rows=11 loops=1)	
2	[...] Join Filter: ((w."wgID" = j."wgID") AND (l.module_no = j.module_no))	
3	[...] Rows Removed by Join Filter: 3992693	
4	[...] -> Nested Loop (cost=0.84..16086.47 rows=124645 width=42) (actual time=0.119..132.801 rows=124772 loops=1)	
5	[...] -> Nested Loop (cost=0.41..7216.99 rows=4511 width=58) (actual time=0.090..52.337 rows=4551 loops=1)	
6	[...] -> Seq Scan on "Student" s (cost=0.00..1205.98 rows=4511 width=12) (actual time=0.046..6.183 rows=4551 loops=1)	
7	[...] Filter: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))	
8	[...] Rows Removed by Filter: 49381	
9	[...] -> Index Scan using "Person_pkey" on "Person" p (cost=0.41..1.33 rows=1 width=46) (actual time=0.010..0.010 rows=1 loops=4551)	
10	[...] Index Cond: ((amka)::text = (s.amka)::text)	
11	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..1.69 rows=28 width=20) (actual time=0.012..0.015 rows=27 loops=4551)	
12	[...] Index Cond: (amka = (p.amka)::text)	
13	[...] Heap Fetches: 0	
14	[...] -> Materialize (cost=0.00..9027.90 rows=54 width=12) (actual time=0.000..0.002 rows=32 loops=124772)	

15	[...] -> Nested Loop (cost=0.00..9027.63 rows=54 width=12) (actual time=57.452..64.860 rows=32 loops=1)
16	[...] Join Filter: (w.module_no = l.module_no)
17	[...] Rows Removed by Join Filter: 72073
18	[...] -> Seq Scan on "LabModule" l (cost=0.00..45.40 rows=345 width=4) (actual time=0.009..0.235 rows=345 loops=1)
19	[...] Filter: (type = 'project'::labmodule_type)
20	[...] Rows Removed by Filter: 1527
21	[...] -> Materialize (cost=0.00..7471.86 rows=292 width=8) (actual time=0.167..0.174 rows=209 loops=345)
22	[...] -> Seq Scan on "Workgroup" w (cost=0.00..7470.40 rows=292 width=8) (actual time=57.434..57.476 rows=209 loops=1)
23	[...] Filter: (grade > '8'::numeric)
24	[...] Rows Removed by Filter: 376063
25	Planning Time: 0.868 ms
26	Execution Time: 602.232 ms

ΜΕ ΕΥΡΕΤΗΡΙΑ, ΧΩΡΙΣ hashjoin	
Εκτελέσεις	Χρόνος Απόκρισης (ms)
1 ^η	314.449
2 ^η	306.388
3 ^η	332.275
ΜΟ	317.704

Όπως και πριν, έχουμε αύξηση του χρόνου εκτέλεσης, αυτή τη φορά κατά 201% σε σχέση με την εκτέλεση με τον συνδυασμό των 3 ευρετηρίων και τη χρήση του hashjoin.

	QUERY PLAN text
1	Nested Loop (cost=28902.03..29855.57 rows=1 width=34) (actual time=269.029..311.638 rows=11 loops=1)
2	[...] Join Filter: (w.module_no = l.module_no)
3	[...] -> Merge Join (cost=28901.76..29841.23 rows=1 width=42) (actual time=265.297..307.106 rows=76 loops=1)
4	[...] Merge Cond: ((j.module_no = w.module_no) AND (j."wgID" = w."wgID"))
5	[...] -> Sort (cost=28890.29..29203.29 rows=125198 width=42) (actual time=264.008..297.278 rows=124772 loops=1)
6	[...] Sort Key: j.module_no, j."wgID"
7	[...] Sort Method: external merge Disk: 6584kB
8	[...] -> Nested Loop (cost=423.91..14436.37 rows=125198 width=42) (actual time=16.947..148.559 rows=124772 loops=1)
9	[...] -> Merge Join (cost=423.48..5527.56 rows=4531 width=58) (actual time=16.904..83.849 rows=4551 loops=1)
10	[...] Merge Cond: ((p.amka)::text = (s.amka)::text)
11	[...] -> Index Scan using "Person_pkey" on "Person" p (cost=0.41..4897.08 rows=55778 width=46) (actual time=0.013..40.720 rows=55623 l...
12	[...] -> Sort (cost=423.07..434.40 rows=4531 width=12) (actual time=16.855..18.020 rows=4551 loops=1)
13	[...] Sort Key: s.amka
14	[...] Sort Method: quicksort Memory: 406kB
15	[...] -> Index Scan using date_idx on "Student" s (cost=0.29..147.91 rows=4531 width=12) (actual time=0.031..0.841 rows=4551 loops=1)
16	[...] Index Cond: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))
17	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..1.69 rows=28 width=20) (actual time=0.009..0.012 rows=27 loops=4551)
18	[...] Index Cond: (amka = (p.amka)::text)
19	[...] Heap Fetches: 0
20	[...] -> Sort (cost=11.46..11.62 rows=64 width=8) (actual time=0.199..0.217 rows=219 loops=1)
21	[...] Sort Key: w.module_no, w."wgID"
22	[...] Sort Method: quicksort Memory: 34kB
23	[...] -> Index Scan using grade_idx on "Workgroup" w (cost=0.42..9.54 rows=64 width=8) (actual time=0.053..0.124 rows=209 loops=1)
24	[...] Index Cond: (grade > '8'::numeric)
25	[...] -> Index Scan using "LabModule_pkey" on "LabModule" l (cost=0.28..14.33 rows=1 width=4) (actual time=0.056..0.059 rows=0 loops=76)
26	[...] Index Cond: (module_no = j.module_no)
27	[...] Filter: (type = 'project'::labmodule_type)
28	[...] Rows Removed by Filter: 1
29	Planning Time: 1.441 ms
30	Execution Time: 314.449 ms

	QUERY PLAN text
1	Nested Loop (cost=28902.03..29855.57 rows=1 width=34) (actual time=262.151..303.360 rows=11 loops=1)
2	[...] Join Filter: (w.module_no = l.module_no)
3	[...] -> Merge Join (cost=28901.76..29841.23 rows=1 width=42) (actual time=258.426..298.580 rows=76 loops=1)
4	[...] Merge Cond: ((j.module_no = w.module_no) AND (j."wglD" = w."wglD"))
5	[...] -> Sort (cost=28890.29..29203.29 rows=125198 width=42) (actual time=256.093..288.801 rows=124772 loops=1)
6	[...] Sort Key: j.module_no, j."wglD"
7	[...] Sort Method: external merge Disk: 6584kB
8	[...] -> Nested Loop (cost=423.91..14436.37 rows=125198 width=42) (actual time=16.799..142.678 rows=124772 loops=1)
9	[...] -> Merge Join (cost=423.48..5527.56 rows=4531 width=58) (actual time=16.768..79.480 rows=4551 loops=1)
10	[...] Merge Cond: ((p.amka)::text = (s.amka)::text)
11	[...] -> Index Scan using "Person_pkey" on "Person" p (cost=0.41..4897.08 rows=55778 width=46) (actual time=0.011..37.023 rows=55623 [...)
12	[...] -> Sort (cost=423.07..434.40 rows=4531 width=12) (actual time=16.734..17.797 rows=4551 loops=1)
13	[...] Sort Key: s.amka
14	[...] Sort Method: quicksort Memory: 406kB

15	[...] -> Index Scan using date_idx on "Student" s (cost=0.29..147.91 rows=4531 width=12) (actual time=0.013..0.705 rows=4551 loops=1)
16	[...] Index Cond: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))
17	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..1.69 rows=28 width=20) (actual time=0.008..0.012 rows=27 loops=4551)
18	[...] Index Cond: (amka = (p.amka)::text)
19	[...] Heap Fetches: 0
20	[...] -> Sort (cost=11.46..11.62 rows=64 width=8) (actual time=0.160..0.177 rows=219 loops=1)
21	[...] Sort Key: w.module_no, w."wglD"
22	[...] Sort Method: quicksort Memory: 34kB
23	[...] -> Index Scan using grade_idx on "Workgroup" w (cost=0.42..9.54 rows=64 width=8) (actual time=0.052..0.095 rows=209 loops=1)
24	[...] Index Cond: (grade > '8'::numeric)
25	[...] -> Index Scan using "LabModule_pkey" on "LabModule" l (cost=0.28..14.33 rows=1 width=4) (actual time=0.059..0.062 rows=0 loops=76)
26	[...] Index Cond: (module_no = j.module_no)
27	[...] Filter: (type = 'project'::labmodule_type)
28	[...] Rows Removed by Filter: 1
29	Planning Time: 1.332 ms
30	Execution Time: 306.388 ms

	QUERY PLAN text
1	Nested Loop (cost=28902.03..29855.57 rows=1 width=34) (actual time=287.429..330.264 rows=11 loops=1)
2	[...] Join Filter: (w.module_no = l.module_no)
3	[...] -> Merge Join (cost=28901.76..29841.23 rows=1 width=42) (actual time=283.701..325.677 rows=76 loops=1)
4	[...] Merge Cond: ((j.module_no = w.module_no) AND (j."wglD" = w."wglD"))
5	[...] -> Sort (cost=28890.29..29203.29 rows=125198 width=42) (actual time=282.472..315.926 rows=124772 loops=1)
6	[...] Sort Key: j.module_no, j."wglD"
7	[...] Sort Method: external merge Disk: 6584kB
8	[...] -> Nested Loop (cost=423.91..14436.37 rows=125198 width=42) (actual time=24.113..163.827 rows=124772 loops=1)
9	[...] -> Merge Join (cost=423.48..5527.56 rows=4531 width=58) (actual time=24.056..97.921 rows=4551 loops=1)
10	[...] Merge Cond: ((p.amka)::text = (s.amka)::text)
11	[...] -> Index Scan using "Person_pkey" on "Person" p (cost=0.41..4897.08 rows=55778 width=46) (actual time=0.013..47.239 rows=55623 [...)
12	[...] -> Sort (cost=423.07..434.40 rows=4531 width=12) (actual time=23.991..25.219 rows=4551 loops=1)
13	[...] Sort Key: s.amka

15	[...] -> Index Scan using date_idx on "Student" s (cost=0.29..147.91 rows=4531 width=12) (actual time=0.022..0.742 rows=4551 loops=1)
16	[...] Index Cond: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))
17	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..1.69 rows=28 width=20) (actual time=0.009..0.012 rows=27 loops=4551)
18	[...] Index Cond: (amka = (p.amka)::text)
19	[...] Heap Fetches: 0
20	[...] -> Sort (cost=11.46..11.62 rows=64 width=8) (actual time=0.156..0.180 rows=219 loops=1)
21	[...] Sort Key: w.module_no, w.wgID"
22	[...] Sort Method: quicksort Memory: 34kB
23	[...] -> Index Scan using grade_idx on "Workgroup" w (cost=0.42..9.54 rows=64 width=8) (actual time=0.049..0.090 rows=209 loops=1)
24	[...] Index Cond: (grade > '8'::numeric)
25	[...] -> Index Scan using "LabModule_pkey" on "LabModule" l (cost=0.28..14.33 rows=1 width=4) (actual time=0.057..0.060 rows=0 loops=76)
26	[...] Index Cond: (module_no = j.module_no)
27	[...] Filter: (type = 'project'::labmodule_type)
28	[...] Rows Removed by Filter: 1
29	Planning Time: 1.530 ms
30	Execution Time: 332.275 ms

ΜΕ ΕΥΡΕΤΗΡΙΑ, ΧΩΡΙΣ hashjoin, ΧΩΡΙΣ mergejoin	
Εκτελέσεις	Χρόνος Απόκρισης (ms)
1 ^η	535.749
2 ^η	526.696
3 ^η	529.398
ΜΟ	530.614

Εδώ ο χρόνος εκτέλεσης αυξάνεται πολύ περισσότερο και συγκεκριμένα κατά 403% σε σχέση με την εκτέλεση με χρήση ευρετηρίων και τα enable_hashjoin και enable_mergejoin on.

QUERY PLAN		
	text	
1	Nested Loop (cost=16.23..73527.36 rows=1 width=34) (actual time=29.600..535.622 rows=11 loops=1)	
2	[...] Join Filter: ((w.wgID" = j.wgID") AND (l.module_no = j.module_no))	
3	[...] Rows Removed by Join Filter: 3992693	
4	[...] -> Nested Loop (cost=1.13..15565.95 rows=130779 width=42) (actual time=0.099..127.130 rows=124772 loops=1)	
5	[...] -> Nested Loop (cost=0.70..6259.97 rows=4733 width=58) (actual time=0.072..47.441 rows=4551 loops=1)	
6	[...] -> Index Scan using date_idx on "Student" s (cost=0.29..152.95 rows=4733 width=12) (actual time=0.028..1.403 rows=4551 loops=1)	
7	[...] Index Cond: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))	
8	[...] -> Index Scan using "Person_pkey" on "Person" p (cost=0.41..1.29 rows=1 width=46) (actual time=0.010..0.010 rows=1 loops=4551)	
9	[...] Index Cond: ((amka)::text = (s.amka)::text)	
10	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..1.69 rows=28 width=20) (actual time=0.012..0.015 rows=27 loops=4551)	
11	[...] Index Cond: (amka = (p.amka)::text)	
12	[...] Heap Fetches: 0	
13	[...] -> Materialize (cost=15.10..745.66 rows=25 width=12) (actual time=0.000..0.001 rows=32 loops=124772)	
14	[...] -> Nested Loop (cost=15.10..745.54 rows=25 width=12) (actual time=0.074..7.459 rows=32 loops=1)	

15	[...] Join Filter: (w.module_no = l.module_no)
16	[...] Rows Removed by Join Filter: 72073
17	[...] -> Bitmap Heap Scan on "LabModule" l (cost=14.67..40.99 rows=345 width=4) (actual time=0.036..0.178 rows=345 loops=1)
18	[...] Recheck Cond: (type = 'project':labmodule_type)
19	[...] Heap Blocks: exact=22
20	[...] -> Bitmap Index Scan on type_idx (cost=0.00..14.59 rows=345 width=0) (actual time=0.024..0.025 rows=345 loops=1)
21	[...] Index Cond: (type = 'project':labmodule_type)
22	[...] -> Materialize (cost=0.42..11.44 rows=134 width=8) (actual time=0.000..0.007 rows=209 loops=345)
23	[...] -> Index Scan using grade_idx on "Workgroup" w (cost=0.42..10.77 rows=134 width=8) (actual time=0.033..0.088 rows=209 loops=1)
24	[...] Index Cond: (grade > '8':numeric)
25	Planning Time: 1.329 ms
26	Execution Time: 535.749 ms

QUERY PLAN		
	text	
1	Nested Loop (cost=16.23..73527.36 rows=1 width=34) (actual time=30.114..526.546 rows=11 loops=1)	
2	[...] Join Filter: ((w."wgID" = j."wgID") AND (l.module_no = j.module_no))	
3	[...] Rows Removed by Join Filter: 3992693	
4	[...] -> Nested Loop (cost=1.13..15565.95 rows=130779 width=42) (actual time=0.118..121.825 rows=124772 loops=1)	
5	[...] -> Nested Loop (cost=0.70..6259.97 rows=4733 width=58) (actual time=0.085..45.183 rows=4551 loops=1)	
6	[...] -> Index Scan using date_idx on "Student" s (cost=0.29..152.95 rows=4733 width=12) (actual time=0.040..1.288 rows=4551 loops=1)	
7	[...] Index Cond: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))	
8	[...] -> Index Scan using "Person_pkey" on "Person" p (cost=0.41..1.29 rows=1 width=46) (actual time=0.009..0.009 rows=1 loops=4551)	
9	[...] Index Cond: ((amka)::text = (s.amka)::text)	
10	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..1.69 rows=28 width=20) (actual time=0.011..0.014 rows=27 loops=4551)	
11	[...] Index Cond: (amka = (p.amka)::text)	
12	[...] Heap Fetches: 0	
13	[...] -> Materialize (cost=15.10..745.66 rows=25 width=12) (actual time=0.000..0.001 rows=32 loops=124772)	
14	[...] -> Nested Loop (cost=15.10..745.54 rows=25 width=12) (actual time=0.077..7.878 rows=32 loops=1)	

15	[...] Join Filter: (w.module_no = l.module_no)
16	[...] Rows Removed by Join Filter: 72073
17	[...] -> Bitmap Heap Scan on "LabModule" l (cost=14.67..40.99 rows=345 width=4) (actual time=0.039..0.215 rows=345 loops=1)
18	[...] Recheck Cond: (type = 'project':labmodule_type)
19	[...] Heap Blocks: exact=22
20	[...] -> Bitmap Index Scan on type_idx (cost=0.00..14.59 rows=345 width=0) (actual time=0.026..0.027 rows=345 loops=1)
21	[...] Index Cond: (type = 'project':labmodule_type)
22	[...] -> Materialize (cost=0.42..11.44 rows=134 width=8) (actual time=0.000..0.008 rows=209 loops=345)
23	[...] -> Index Scan using grade_idx on "Workgroup" w (cost=0.42..10.77 rows=134 width=8) (actual time=0.033..0.091 rows=209 loops=1)
24	[...] Index Cond: (grade > '8':numeric)
25	Planning Time: 0.991 ms
26	Execution Time: 526.696 ms

QUERY PLAN		text	
1	Nested Loop	(cost=16.23..73527.36 rows=1 width=34) (actual time=29.826..529.309 rows=11 loops=1)	
2	[...]	Join Filter: ((w."wgID" = j."wgID") AND (l.module_no = j.module_no))	
3	[...]	Rows Removed by Join Filter: 3992693	
4	[...]-> Nested Loop	(cost=1.13..15565.95 rows=130779 width=42) (actual time=0.067..126.748 rows=124772 loops=1)	
5	[...]-> Nested Loop	(cost=0.70..6259.97 rows=4733 width=58) (actual time=0.046..46.828 rows=4551 loops=1)	
6	[...]-> Index Scan using date_idx on "Student" s	(cost=0.29..152.95 rows=4733 width=12) (actual time=0.013..1.348 rows=4551 loops=1)	
7	[...]	Index Cond: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))	
8	[...]-> Index Scan using "Person_pkey" on "Person" p	(cost=0.41..1.29 rows=1 width=46) (actual time=0.010..0.010 rows=1 loops=4551)	
9	[...]	Index Cond: ((amka)::text = (s.amka)::text)	
10	[...]-> Index Only Scan using "Joins_pkey" on "Joins" j	(cost=0.43..1.69 rows=28 width=20) (actual time=0.012..0.015 rows=27 loops=4551)	
11	[...]	Index Cond: (amka = (p.amka)::text)	
12	[...]	Heap Fetches: 0	
13	[...]-> Materialize	(cost=15.10..745.66 rows=25 width=12) (actual time=0.000..0.001 rows=32 loops=124772)	
14	[...]-> Nested Loop	(cost=15.10..745.54 rows=25 width=12) (actual time=0.054..7.596 rows=32 loops=1)	
15	[...]	Join Filter: (w.module_no = l.module_no)	
16	[...]	Rows Removed by Join Filter: 72073	
17	[...]-> Bitmap Heap Scan on "LabModule" l	(cost=14.67..40.99 rows=345 width=4) (actual time=0.029..0.178 rows=345 loops=1)	
18	[...]	Recheck Cond: (type = 'project'::labmodule_type)	
19	[...]	Heap Blocks: exact=22	
20	[...]-> Bitmap Index Scan on type_idx	(cost=0.00..14.59 rows=345 width=0) (actual time=0.019..0.020 rows=345 loops=1)	
21	[...]	Index Cond: (type = 'project'::labmodule_type)	
22	[...]-> Materialize	(cost=0.42..11.44 rows=134 width=8) (actual time=0.000..0.008 rows=209 loops=345)	
23	[...]-> Index Scan using grade_idx on "Workgroup" w	(cost=0.42..10.77 rows=134 width=8) (actual time=0.022..0.062 rows=209 loops=1)	
24	[...]	Index Cond: (grade > '8'::numeric)	
25	Planning Time: 0.788 ms		
26	Execution Time: 529.398 ms		

Τέλος, εναλλάσσοντας την σειρά των συνδέσεων (join) καταλήξαμε ότι η απόδοση του query έπεφτε. Οπότε, συμπεραίνουμε ότι η αρχική σειρά των συνδέσεων είναι η καλύτερη δυνατή. Ενδεικτικά, παραθέτουμε τα στιγμιότυπα από την εξής εκτέλεση, είτε με ευρετήρια, είτε χωρίς:

EXPLAIN ANALYSE

```
SELECT name, surname
FROM "Workgroup" w JOIN "Joins" j ON (w."wgID" = j."wgID")
JOIN "LabModule" l ON (l.module_no = w.module_no) AND (l.module_no = j.module_no)
JOIN "Student" s ON (j.amka = s.amka)
JOIN "Person" p ON (s.amka = p.amka)
WHERE s.entry_date > '2010-09-01'::date
AND s.entry_date < '2011-09-01'::date
AND l.type = 'project'
AND w.grade > 8
```

ΧΩΡΙΣ ΕΥΡΕΤΗΡΙΑ	
Εκτελέσεις	Χρόνος Απόκρισης (ms)
1 ^η	171.492
2 ^η	185.681
3 ^η	179.695
ΜΟ	178.956

Παρατηρούμε μικρή αύξηση του χρόνου κατά 4%.

QUERY PLAN	
	text
1	Hash Join (cost=8785.82..20514.39 rows=1 width=34) (actual time=72.526..171.407 rows=11 loops=1)
2	[...] Hash Cond: ((j."wglID" = w."wglID") AND (j.module_no = l.module_no))
3	[...] -> Nested Loop (cost=1263.39..12047.17 rows=125971 width=42) (actual time=6.014..103.720 rows=124772 loops=1)
4	[...] Join Filter: ((s.amka)::text = (j.amka)::text)
5	[...] -> Hash Join (cost=1262.97..2764.18 rows=4559 width=58) (actual time=5.902..24.306 rows=4551 loops=1)
6	[...] Hash Cond: ((p.amka)::text = (s.amka)::text)
7	[...] -> Seq Scan on "Person" p (cost=0.00..1354.78 rows=55778 width=46) (actual time=0.026..3.690 rows=55778 loops=1)
8	[...] -> Hash (cost=1205.98..1205.98 rows=4559 width=12) (actual time=5.841..5.842 rows=4551 loops=1)
9	[...] Buckets: 8192 Batches: 1 Memory Usage: 260kB
10	[...] -> Seq Scan on "Student" s (cost=0.00..1205.98 rows=4559 width=12) (actual time=0.025..5.283 rows=4551 loops=1)
11	[...] Filter: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))
12	[...] Rows Removed by Filter: 49381
13	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..1.69 rows=28 width=20) (actual time=0.011..0.014 rows=27 loops=4551)
14	[...] Index Cond: (amka = (p.amka)::text)
15	[...] Heap Fetches: 0
16	[...] -> Hash (cost=7521.66..7521.66 rows=51 width=12) (actual time=58.750..58.753 rows=32 loops=1)
17	[...] Buckets: 1024 Batches: 1 Memory Usage: 10kB
18	[...] -> Hash Join (cost=49.71..7521.66 rows=51 width=12) (actual time=58.678..58.739 rows=32 loops=1)
19	[...] Hash Cond: (w.module_no = l.module_no)
20	[...] -> Seq Scan on "Workgroup" w (cost=0.00..7470.40 rows=278 width=8) (actual time=58.385..58.424 rows=209 loops=1)
21	[...] Filter: (grade > '8'::numeric)
22	[...] Rows Removed by Filter: 376063
23	[...] -> Hash (cost=45.40..45.40 rows=345 width=4) (actual time=0.271..0.272 rows=345 loops=1)
24	[...] Buckets: 1024 Batches: 1 Memory Usage: 21kB
25	[...] -> Seq Scan on "LabModule" l (cost=0.00..45.40 rows=345 width=4) (actual time=0.038..0.230 rows=345 loops=1)
26	[...] Filter: (type = 'project'::labmodule_type)
27	[...] Rows Removed by Filter: 1527
28	Planning Time: 1.323 ms
29	Execution Time: 171.492 ms

	QUERY PLAN	
	text	
1	Hash Join (cost=8785.82..20514.39 rows=1 width=34) (actual time=81.613..185.587 rows=11 loops=1)	
2	[...] Hash Cond: ((j."wglD" = w."wglD") AND (j.module_no = l.module_no))	
3	[...] -> Nested Loop (cost=1263.39..12047.17 rows=125971 width=42) (actual time=7.236..110.855 rows=124772 loops=1)	
4	[...] Join Filter: ((s.amka)::text = (j.amka)::text)	
5	[...] -> Hash Join (cost=1262.97..2764.18 rows=4559 width=58) (actual time=7.160..27.850 rows=4551 loops=1)	
6	[...] Hash Cond: ((p.amka)::text = (s.amka)::text)	
7	[...] -> Seq Scan on "Person" p (cost=0.00..1354.78 rows=55778 width=46) (actual time=0.029..3.792 rows=55778 loops=1)	
8	[...] -> Hash (cost=1205.98..1205.98 rows=4559 width=12) (actual time=7.096..7.097 rows=4551 loops=1)	
9	[...] Buckets: 8192 Batches: 1 Memory Usage: 260kB	
10	[...] -> Seq Scan on "Student" s (cost=0.00..1205.98 rows=4559 width=12) (actual time=0.042..6.472 rows=4551 loops=1)	
11	[...] Filter: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))	
12	[...] Rows Removed by Filter: 49381	
13	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..1.69 rows=28 width=20) (actual time=0.012..0.015 rows=27 loops=4551)	
14	[...] Index Cond: (amka = (p.amka)::text)	

15	[...] Heap Fetches: 0
16	[...] -> Hash (cost=7521.66..7521.66 rows=51 width=12) (actual time=65.453..65.458 rows=32 loops=1)
17	[...] Buckets: 1024 Batches: 1 Memory Usage: 10kB
18	[...] -> Hash Join (cost=49.71..7521.66 rows=51 width=12) (actual time=65.381..65.445 rows=32 loops=1)
19	[...] Hash Cond: (w.module_no = l.module_no)
20	[...] -> Seq Scan on "Workgroup" w (cost=0.00..7470.40 rows=278 width=8) (actual time=65.044..65.086 rows=209 loops=1)
21	[...] Filter: (grade > '8'::numeric)
22	[...] Rows Removed by Filter: 376063
23	[...] -> Hash (cost=45.40..45.40 rows=345 width=4) (actual time=0.309..0.311 rows=345 loops=1)
24	[...] Buckets: 1024 Batches: 1 Memory Usage: 21kB
25	[...] -> Seq Scan on "LabModule" l (cost=0.00..45.40 rows=345 width=4) (actual time=0.040..0.265 rows=345 loops=1)
26	[...] Filter: (type = 'project'::labmodule_type)
27	[...] Rows Removed by Filter: 1527
28	Planning Time: 1.998 ms
29	Execution Time: 185.681 ms

	QUERY PLAN	
	text	
1	Hash Join (cost=8785.82..20514.39 rows=1 width=34) (actual time=76.560..179.618 rows=11 loops=1)	
2	[...] Hash Cond: ((j."wglD" = w."wglD") AND (j.module_no = l.module_no))	
3	[...] -> Nested Loop (cost=1263.39..12047.17 rows=125971 width=42) (actual time=6.018..109.053 rows=124772 loops=1)	
4	[...] Join Filter: ((s.amka)::text = (j.amka)::text)	
5	[...] -> Hash Join (cost=1262.97..2764.18 rows=4559 width=58) (actual time=5.947..26.552 rows=4551 loops=1)	
6	[...] Hash Cond: ((p.amka)::text = (s.amka)::text)	
7	[...] -> Seq Scan on "Person" p (cost=0.00..1354.78 rows=55778 width=46) (actual time=0.023..3.854 rows=55778 loops=1)	
8	[...] -> Hash (cost=1205.98..1205.98 rows=4559 width=12) (actual time=5.898..5.900 rows=4551 loops=1)	
9	[...] Buckets: 8192 Batches: 1 Memory Usage: 260kB	
10	[...] -> Seq Scan on "Student" s (cost=0.00..1205.98 rows=4559 width=12) (actual time=0.025..5.361 rows=4551 loops=1)	
11	[...] Filter: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))	
12	[...] Rows Removed by Filter: 49381	
13	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..1.69 rows=28 width=20) (actual time=0.011..0.014 rows=27 loops=4551)	
14	[...] Index Cond: (amka = (p.amka)::text)	

15	[...] Heap Fetches: 0
16	[...] -> Hash (cost=7521.66..7521.66 rows=51 width=12) (actual time=61.401..61.404 rows=32 loops=1)
17	[...] Buckets: 1024 Batches: 1 Memory Usage: 10kB
18	[...] -> Hash Join (cost=49.71..7521.66 rows=51 width=12) (actual time=61.333..61.393 rows=32 loops=1)
19	[...] Hash Cond: (w.module_no = l.module_no)
20	[...] -> Seq Scan on "Workgroup" w (cost=0.00..7470.40 rows=278 width=8) (actual time=61.030..61.069 rows=209 loops=1)
21	[...] Filter: (grade > '8'::numeric)
22	[...] Rows Removed by Filter: 376063
23	[...] -> Hash (cost=45.40..45.40 rows=345 width=4) (actual time=0.280..0.281 rows=345 loops=1)
24	[...] Buckets: 1024 Batches: 1 Memory Usage: 21kB
25	[...] -> Seq Scan on "LabModule" l (cost=0.00..45.40 rows=345 width=4) (actual time=0.038..0.238 rows=345 loops=1)
26	[...] Filter: (type = 'project'::labmodule_type)
27	[...] Rows Removed by Filter: 1527
28	Planning Time: 1.363 ms
29	Execution Time: 179.695 ms

ΜΕ ΕΥΡΕΤΗΡΙΑ	
Εκτελέσεις	Χρόνος Απόκρισης (ms)
1 ^η	121.518
2 ^η	114.783
3 ^η	125.765
ΜΟ	120.690

Παρατηρούμε αύξηση του χρόνου εκτέλεσης περίπου κατά 14%.

QUERY PLAN		
	text	
1	Hash Join (cost=272.70..12360.21 rows=1 width=34) (actual time=10.486..121.396 rows=11 loops=1)	
2	[...] Hash Cond: ((j."wglD" = w."wglD") AND (j.module_no = l.module_no))	
3	[...] -> Nested Loop (cost=212.09..11321.65 rows=130392 width=42) (actual time=1.373..111.937 rows=124772 loops=1)	
4	[...] Join Filter: ((s.amka)::text = (j.amka)::text)	
5	[...] -> Hash Join (cost=211.66..1712.87 rows=4719 width=58) (actual time=1.318..25.235 rows=4551 loops=1)	
6	[...] Hash Cond: ((p.amka)::text = (s.amka)::text)	
7	[...] -> Seq Scan on "Person" p (cost=0.00..1354.78 rows=55778 width=46) (actual time=0.009..3.898 rows=55778 loops=1)	
8	[...] -> Hash (cost=152.67..152.67 rows=4719 width=12) (actual time=1.275..1.276 rows=4551 loops=1)	
9	[...] Buckets: 8192 Batches: 1 Memory Usage: 260kB	
10	[...] -> Index Scan using date_idx on "Student" s (cost=0.29..152.67 rows=4719 width=12) (actual time=0.023..0.679 rows=4551 loops=1)	
11	[...] Index Cond: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))	
12	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..1.69 rows=28 width=20) (actual time=0.012..0.015 rows=27 loops=4551)	
13	[...] Index Cond: (amka = (p.amka)::text)	
14	[...] Heap Fetches: 0	

15	[...] -> Hash (cost=59.99..59.99 rows=42 width=12) (actual time=0.277..0.282 rows=32 loops=1)
16	[...] Buckets: 1024 Batches: 1 Memory Usage: 10kB
17	[...] -> Hash Join (cost=45.72..59.99 rows=42 width=12) (actual time=0.220..0.275 rows=32 loops=1)
18	[...] Hash Cond: (w.module_no = l.module_no)
19	[...] -> Index Scan using grade_idx on "Workgroup" w (cost=0.42..13.41 rows=228 width=8) (actual time=0.033..0.060 rows=209 loops=1)
20	[...] Index Cond: (grade > '8'::numeric)
21	[...] -> Hash (cost=40.99..40.99 rows=345 width=4) (actual time=0.182..0.184 rows=345 loops=1)
22	[...] Buckets: 1024 Batches: 1 Memory Usage: 21kB
23	[...] -> Bitmap Heap Scan on "LabModule" l (cost=14.67..40.99 rows=345 width=4) (actual time=0.030..0.143 rows=345 loops=1)
24	[...] Recheck Cond: (type = 'project'::labmodule_type)
25	[...] Heap Blocks: exact=22
26	[...] -> Bitmap Index Scan on type_idx (cost=0.00..14.59 rows=345 width=0) (actual time=0.019..0.020 rows=345 loops=1)
27	[...] Index Cond: (type = 'project'::labmodule_type)
28	Planning Time: 1.634 ms
29	Execution Time: 121.518 ms

QUERY PLAN

text



1	Hash Join (cost=272.70..12360.21 rows=1 width=34) (actual time=9.421..114.680 rows=11 loops=1)
2	[...] Hash Cond: ((j."wglD" = w."wglD") AND (j.module_no = l.module_no))
3	[...] -> Nested Loop (cost=212.09..11321.65 rows=130392 width=42) (actual time=1.260..105.415 rows=124772 loops=1)
4	[...] Join Filter: ((s.amka)::text = (j.amka)::text)
5	[...] -> Hash Join (cost=211.66..1712.87 rows=4719 width=58) (actual time=1.210..22.469 rows=4551 loops=1)
6	[...] Hash Cond: ((p.amka)::text = (s.amka)::text)
7	[...] -> Seq Scan on "Person" p (cost=0.00..1354.78 rows=55778 width=46) (actual time=0.009..3.728 rows=55778 loops=1)
8	[...] -> Hash (cost=152.67..152.67 rows=4719 width=12) (actual time=1.183..1.185 rows=4551 loops=1)
9	[...] Buckets: 8192 Batches: 1 Memory Usage: 260kB
10	[...] -> Index Scan using date_idx on "Student" s (cost=0.29..152.67 rows=4719 width=12) (actual time=0.010..0.660 rows=4551 loops=1)
11	[...] Index Cond: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))
12	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..1.69 rows=28 width=20) (actual time=0.012..0.015 rows=27 loops=4551)
13	[...] Index Cond: (amka = (p.amka)::text)
14	[...] Heap Fetches: 0
15	[...] -> Hash (cost=59.99..59.99 rows=42 width=12) (actual time=0.266..0.270 rows=32 loops=1)
16	[...] Buckets: 1024 Batches: 1 Memory Usage: 10kB
17	[...] -> Hash Join (cost=45.72..59.99 rows=42 width=12) (actual time=0.208..0.263 rows=32 loops=1)
18	[...] Hash Cond: (w.module_no = l.module_no)
19	[...] -> Index Scan using grade_idx on "Workgroup" w (cost=0.42..13.41 rows=228 width=8) (actual time=0.025..0.052 rows=209 loops=1)
20	[...] Index Cond: (grade > '8'::numeric)
21	[...] -> Hash (cost=40.99..40.99 rows=345 width=4) (actual time=0.179..0.180 rows=345 loops=1)
22	[...] Buckets: 1024 Batches: 1 Memory Usage: 21kB
23	[...] -> Bitmap Heap Scan on "LabModule" l (cost=14.67..40.99 rows=345 width=4) (actual time=0.031..0.143 rows=345 loops=1)
24	[...] Recheck Cond: (type = 'project'::labmodule_type)
25	[...] Heap Blocks: exact=22
26	[...] -> Bitmap Index Scan on type_idx (cost=0.00..14.59 rows=345 width=0) (actual time=0.018..0.019 rows=345 loops=1)
27	[...] Index Cond: (type = 'project'::labmodule_type)
28	Planning Time: 1.444 ms
29	Execution Time: 114.783 ms

	QUERY PLAN	
	text	
1	Hash Join (cost=272.70..12360.21 rows=1 width=34) (actual time=10.769..125.656 rows=11 loops=1)	
2	[...] Hash Cond: ((j."wglD" = w."wglD") AND (j.module_no = l.module_no))	
3	[...] -> Nested Loop (cost=212.09..11321.65 rows=130392 width=42) (actual time=1.381..116.159 rows=124772 loops=1)	
4	[...] Join Filter: ((s.amka)::text = (j.amka)::text)	
5	[...] -> Hash Join (cost=211.66..1712.87 rows=4719 width=58) (actual time=1.343..25.580 rows=4551 loops=1)	
6	[...] Hash Cond: ((p.amka)::text = (s.amka)::text)	
7	[...] -> Seq Scan on "Person" p (cost=0.00..1354.78 rows=55778 width=46) (actual time=0.009..3.851 rows=55778 loops=1)	
8	[...] -> Hash (cost=152.67..152.67 rows=4719 width=12) (actual time=1.316..1.316 rows=4551 loops=1)	
9	[...] Buckets: 8192 Batches: 1 Memory Usage: 260kB	
10	[...] -> Index Scan using date_idx on "Student" s (cost=0.29..152.67 rows=4719 width=12) (actual time=0.011..0.788 rows=4551 loops=1)	
11	[...] Index Cond: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))	
12	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..1.69 rows=28 width=20) (actual time=0.013..0.016 rows=27 loops=4551)	
13	[...] Index Cond: (amka = (p.amka)::text)	
14	[...] Heap Fetches: 0	
15	[...] -> Hash (cost=59.99..59.99 rows=42 width=12) (actual time=0.265..0.269 rows=32 loops=1)	
16	[...] Buckets: 1024 Batches: 1 Memory Usage: 10kB	
17	[...] -> Hash Join (cost=45.72..59.99 rows=42 width=12) (actual time=0.206..0.262 rows=32 loops=1)	
18	[...] Hash Cond: (w.module_no = l.module_no)	
19	[...] -> Index Scan using grade_idx on "Workgroup" w (cost=0.42..13.41 rows=228 width=8) (actual time=0.025..0.053 rows=209 loops=1)	
20	[...] Index Cond: (grade > '8'::numeric)	
21	[...] -> Hash (cost=40.99..40.99 rows=345 width=4) (actual time=0.176..0.177 rows=345 loops=1)	
22	[...] Buckets: 1024 Batches: 1 Memory Usage: 21kB	
23	[...] -> Bitmap Heap Scan on "LabModule" l (cost=14.67..40.99 rows=345 width=4) (actual time=0.029..0.143 rows=345 loops=1)	
24	[...] Recheck Cond: (type = 'project'::labmodule_type)	
25	[...] Heap Blocks: exact=22	
26	[...] -> Bitmap Index Scan on type_idx (cost=0.00..14.59 rows=345 width=0) (actual time=0.018..0.019 rows=345 loops=1)	
27	[...] Index Cond: (type = 'project'::labmodule_type)	
28	Planning Time: 1.830 ms	
29	Execution Time: 125.765 ms	

Βήμα 3:

Αρχικά, χρησιμοποιώντας την υλοποίηση του ζητούμενου 1.4 αυξήσαμε το πλήθος των δεδομένων της βάσης εισάγοντας τυχαίες εργαστηριακές εργασίες και ομάδες εργασίας για όλα τα εξαμηνιαία εργαστηριακά μαθήματα γνωστικών αντικειμένων ΕΝΕ, ΗΡΥ, ΣΥΣ. Συγκεκριμένα, η υπάρχουσα βάση διέθετε 1872 πλειάδες στο LabModule , οι οποίες αυξήθηκαν στις 111872 πλειάδες (ΕΝΕ 5000 εισαγωγές για καθένα από τα 6 μαθήματα , ΗΡΥ 2000 εισαγωγές για καθένα από τα 20 μαθήματα και ΣΥΣ 4000 εισαγωγές για καθένα από τα 10 μαθήματα). Εν συνεχεία, επαναλάβαμε την διαδικασία του βήματος 2.

Έτσι, απενεργοποιήθηκε η δυνατότητα δημιουργίας παράλληλων πλάνων εκτέλεσης, εκτελέστηκε η εντολή Explain Analyze. Ενδεικτικά πάρθηκαν 3 χρόνοι εκτέλεσης και εξήχθη ο μέσος όρος αυτών:

ΧΩΡΙΣ ΕΥΡΕΤΗΡΙΑ	
Εκτελέσεις	Χρόνος Απόκρισης (ms)
1 ^η	3413.551
2 ^η	3335.660
3 ^η	3349.222
ΜΟ	3366.144

Πιο αναλυτικά, γίνεται sequential scan και inner join μεταξύ των πινάκων. Τα στιγμιότυπα από το πλάνο εκτέλεσης για 3 ενδεικτικές εκτελέσεις φαίνονται παρακάτω:

	QUERY PLAN text
1	Nested Loop (cost=12999.97..31494.87 rows=1 width=34) (actual time=108.362..3413.212 rows=172 loops=1)
2	[...] Join Filter: (w.module_no = l.module_no)
3	[...] -> Hash Join (cost=12999.55..30655.38 rows=1 width=42) (actual time=105.938..266.550 rows=789 loops=1)
4	[...] Hash Cond: ((j."wglD" = w."wglD") AND (j.module_no = w.module_no))
5	[...] -> Nested Loop (cost=1262.55..16764.42 rows=172316 width=42) (actual time=6.063..123.404 rows=132558 loops=1)
6	[...] -> Hash Join (cost=1262.12..2763.33 rows=4491 width=58) (actual time=5.980..32.580 rows=4551 loops=1)
7	[...] Hash Cond: ((p.amka)::text = (s.amka)::text)
8	[...] -> Seq Scan on "Person" p (cost=0.00..1354.78 rows=55778 width=46) (actual time=0.027..4.392 rows=55778 loops=1)
9	[...] -> Hash (cost=1205.98..1205.98 rows=4491 width=12) (actual time=5.914..5.915 rows=4551 loops=1)
10	[...] Buckets: 8192 Batches: 1 Memory Usage: 260kB
11	[...] -> Seq Scan on "Student" s (cost=0.00..1205.98 rows=4491 width=12) (actual time=0.025..5.296 rows=4551 loops=1)
12	[...] Filter: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))
13	[...] Rows Removed by Filter: 49381
14	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..2.53 rows=59 width=20) (actual time=0.013..0.017 rows=29 loops=4551)
15	[...] Index Cond: (amka = (p.amka)::text)
16	[...] Heap Fetches: 0
17	[...] -> Hash (cost=11441.40..11441.40 rows=19707 width=8) (actual time=99.562..99.563 rows=19132 loops=1)
18	[...] Buckets: 32768 Batches: 1 Memory Usage: 1004kB
19	[...] -> Seq Scan on "Workgroup" w (cost=0.00..11441.40 rows=19707 width=8) (actual time=0.069..94.838 rows=19132 loops=1)
20	[...] Filter: (grade > '8'::numeric)
21	[...] Rows Removed by Filter: 557140
22	[...] -> Index Scan using "LabModule_pkey" on "LabModule" l (cost=0.42..839.47 rows=1 width=4) (actual time=3.629..3.984 rows=0 loops=789)
23	[...] Index Cond: (module_no = j.module_no)
24	[...] Filter: (type = 'project'::labmodule_type)
25	[...] Rows Removed by Filter: 1
26	Planning Time: 2.132 ms
27	Execution Time: 3413.551 ms

	QUERY PLAN text
1	Nested Loop (cost=12999.97..31494.87 rows=1 width=34) (actual time=106.958..3335.345 rows=172 loops=1)
2	[...] Join Filter: (w.module_no = l.module_no)
3	[...] -> Hash Join (cost=12999.55..30655.38 rows=1 width=42) (actual time=104.513..252.150 rows=789 loops=1)
4	[...] Hash Cond: ((j."wglD" = w."wglD") AND (j.module_no = w.module_no))
5	[...] -> Nested Loop (cost=1262.55..16764.42 rows=172316 width=42) (actual time=6.456..115.565 rows=132558 loops=1)
6	[...] -> Hash Join (cost=1262.12..2763.33 rows=4491 width=58) (actual time=6.386..30.311 rows=4551 loops=1)
7	[...] Hash Cond: ((p.amka)::text = (s.amka)::text)
8	[...] -> Seq Scan on "Person" p (cost=0.00..1354.78 rows=55778 width=46) (actual time=0.042..4.344 rows=55778 loops=1)
9	[...] -> Hash (cost=1205.98..1205.98 rows=4491 width=12) (actual time=6.272..6.274 rows=4551 loops=1)
10	[...] Buckets: 8192 Batches: 1 Memory Usage: 260kB
11	[...] -> Seq Scan on "Student" s (cost=0.00..1205.98 rows=4491 width=12) (actual time=0.031..5.638 rows=4551 loops=1)
12	[...] Filter: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))
13	[...] Rows Removed by Filter: 49381
14	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..2.53 rows=59 width=20) (actual time=0.012..0.016 rows=29 loops=4551)
15	[...] Index Cond: (amka = (p.amka)::text)
16	[...] Heap Fetches: 0
17	[...] -> Hash (cost=11441.40..11441.40 rows=19707 width=8) (actual time=97.672..97.672 rows=19132 loops=1)
18	[...] Buckets: 32768 Batches: 1 Memory Usage: 1004kB
19	[...] -> Seq Scan on "Workgroup" w (cost=0.00..11441.40 rows=19707 width=8) (actual time=0.062..92.511 rows=19132 loops=1)
20	[...] Filter: (grade > '8'::numeric)
21	[...] Rows Removed by Filter: 557140
22	[...] -> Index Scan using "LabModule_pkey" on "LabModule" l (cost=0.42..839.47 rows=1 width=4) (actual time=3.555..3.904 rows=0 loops=789)
23	[...] Index Cond: (module_no = j.module_no)
24	[...] Filter: (type = 'project'::labmodule_type)
25	[...] Rows Removed by Filter: 1
26	Planning Time: 1.736 ms
27	Execution Time: 3335.660 ms

	QUERY PLAN text
1	Nested Loop (cost=12999.97..31494.87 rows=1 width=34) (actual time=108.150..3348.915 rows=172 loops=1)
2	[...] Join Filter: (w.module_no = l.module_no)
3	[...] -> Hash Join (cost=12999.55..30655.38 rows=1 width=42) (actual time=105.591..254.020 rows=789 loops=1)
4	[...] Hash Cond: ((j."wglD" = w."wglD") AND (j.module_no = w.module_no))
5	[...] -> Nested Loop (cost=1262.55..16764.42 rows=172316 width=42) (actual time=6.230..115.629 rows=132558 loops=1)
6	[...] -> Hash Join (cost=1262.12..2763.33 rows=4491 width=58) (actual time=6.176..30.138 rows=4551 loops=1)
7	[...] Hash Cond: ((p.amka)::text = (s.amka)::text)
8	[...] -> Seq Scan on "Person" p (cost=0.00..1354.78 rows=55778 width=46) (actual time=0.031..4.261 rows=55778 loops=1)
9	[...] -> Hash (cost=1205.98..1205.98 rows=4491 width=12) (actual time=6.072..6.074 rows=4551 loops=1)
10	[...] Buckets: 8192 Batches: 1 Memory Usage: 260kB
11	[...] -> Seq Scan on "Student" s (cost=0.00..1205.98 rows=4491 width=12) (actual time=0.028..5.454 rows=4551 loops=1)
12	[...] Filter: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))
13	[...] Rows Removed by Filter: 49381
14	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..2.53 rows=59 width=20) (actual time=0.012..0.016 rows=29 loops=4551)

15	[...] Index Cond: (amka = (p.amka)::text)
16	[...] Heap Fetches: 0
17	[...] -> Hash (cost=11441.40..11441.40 rows=19707 width=8) (actual time=99.062..99.063 rows=19132 loops=1)
18	[...] Buckets: 32768 Batches: 1 Memory Usage: 1004kB
19	[...] -> Seq Scan on "Workgroup" w (cost=0.00..11441.40 rows=19707 width=8) (actual time=0.065..93.873 rows=19132 loops=1)
20	[...] Filter: (grade > '8'::numeric)
21	[...] Rows Removed by Filter: 557140
22	[...] -> Index Scan using "LabModule_pkey" on "LabModule" l (cost=0.42..839.47 rows=1 width=4) (actual time=3.569..3.919 rows=0 loops=789)
23	[...] Index Cond: (module_no = j.module_no)
24	[...] Filter: (type = 'project'::labmodule_type)
25	[...] Rows Removed by Filter: 1
26	Planning Time: 2.406 ms
27	Execution Time: 3349.222 ms

Εν συνεχεία, μετά από εκτελέσεις με χρήση διαφόρων ευρετηρίων καταλήξαμε ότι το μοναδικό που βελτιώνει την απόδοση είναι τύπου btree για την τιμή grade.

ΜΕ ΕΥΡΕΤΗΡΙΟ ΤΥΠΟΥ BTREE (grade)	
Εκτελέσεις	Χρόνος Απόκρισης (ms)
1 ^η	2702.339
2 ^η	2696.882
3 ^η	2705.798
ΜΟ	2701.673

Παρατηρούμε ότι με τη χρήση του ευρετηρίου Grade_idx υπάρχει αισθητή μείωση του μέσου χρόνου εκτέλεσης περίπου κατά 20%, γεγονός που συνάδει με τη θεωρία καθώς η συνθήκη που αφορά στον βαθμό έχει μεγάλο selectivity.

	QUERY PLAN	
	text	🔒
1	Nested Loop (cost=2224.44..21167.06 rows=1 width=34) (actual time=14.529..2702.083 rows=172 loops=1)	
2	[...] Join Filter: (w.module_no = l.module_no)	
3	[...] -> Hash Join (cost=2224.03..20327.57 rows=1 width=42) (actual time=12.180..108.557 rows=789 loops=1)	
4	[...] Hash Cond: ((j."wglID" = w."wglID") AND (j.module_no = w.module_no))	
5	[...] -> Nested Loop (cost=1261.17..17263.52 rows=168095 width=42) (actual time=5.662..78.364 rows=132558 loops=1)	
6	[...] -> Hash Join (cost=1260.74..2761.95 rows=4381 width=58) (actual time=5.638..20.136 rows=4551 loops=1)	
7	[...] Hash Cond: ((p.amka)::text = (s.amka)::text)	
8	[...] -> Seq Scan on "Person" p (cost=0.00..1354.78 rows=55778 width=46) (actual time=0.029..3.399 rows=55778 loops=1)	
9	[...] -> Hash (cost=1205.98..1205.98 rows=4381 width=12) (actual time=5.571..5.572 rows=4551 loops=1)	
10	[...] Buckets: 8192 Batches: 1 Memory Usage: 260kB	
11	[...] -> Seq Scan on "Student" s (cost=0.00..1205.98 rows=4381 width=12) (actual time=0.026..5.090 rows=4551 loops=1)	
12	[...] Filter: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))	
13	[...] Rows Removed by Filter: 49381	
14	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..2.73 rows=58 width=20) (actual time=0.008..0.011 rows=29 loops=4551)	

15	[...] Index Cond: (amka = (p.amka)::text)
16	[...] Heap Fetches: 0
17	[...] -> Hash (cost=680.19..680.19 rows=18844 width=8) (actual time=6.295..6.296 rows=19132 loops=1)
18	[...] Buckets: 32768 Batches: 1 Memory Usage: 1004kB
19	[...] -> Index Scan using grade_idx on "Workgroup" w (cost=0.42..680.19 rows=18844 width=8) (actual time=0.027..3.583 rows=19132 loops=1)
20	[...] Index Cond: (grade > '8'::numeric)
21	[...] -> Index Scan using "LabModule_pkey" on "LabModule" l (cost=0.42..839.47 rows=1 width=4) (actual time=2.992..3.286 rows=0 loops=789)
22	[...] Index Cond: (module_no = j.module_no)
23	[...] Filter: (type = 'project'::labmodule_type)
24	[...] Rows Removed by Filter: 1
25	Planning Time: 1.451 ms
26	Execution Time: 2702.339 ms

QUERY PLAN		
	text	
1	Nested Loop (cost=2224.44..21167.06 rows=1 width=34) (actual time=13.207..2696.632 rows=172 loops=1)	
2	[...] Join Filter: (w.module_no = l.module_no)	
3	[...] -> Hash Join (cost=2224.03..20327.57 rows=1 width=42) (actual time=11.286..111.894 rows=789 loops=1)	
4	[...] Hash Cond: ((j."wgID" = w."wgID") AND (j.module_no = w.module_no))	
5	[...] -> Nested Loop (cost=1261.17..17263.52 rows=168095 width=42) (actual time=4.772..82.297 rows=132558 loops=1)	
6	[...] -> Hash Join (cost=1260.74..2761.95 rows=4381 width=58) (actual time=4.748..19.506 rows=4551 loops=1)	
7	[...] Hash Cond: ((p.amka)::text = (s.amka)::text)	
8	[...] -> Seq Scan on "Person" p (cost=0.00..1354.78 rows=55778 width=46) (actual time=0.018..3.310 rows=55778 loops=1)	
9	[...] -> Hash (cost=1205.98..1205.98 rows=4381 width=12) (actual time=4.696..4.697 rows=4551 loops=1)	
10	[...] Buckets: 8192 Batches: 1 Memory Usage: 260kB	
11	[...] -> Seq Scan on "Student" s (cost=0.00..1205.98 rows=4381 width=12) (actual time=0.021..4.157 rows=4551 loops=1)	
12	[...] Filter: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))	
13	[...] Rows Removed by Filter: 49381	
14	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..2.73 rows=58 width=20) (actual time=0.008..0.011 rows=29 loops=4551)	

15	[...] Index Cond: (amka = (p.amka)::text)
16	[...] Heap Fetches: 0
17	[...] -> Hash (cost=680.19..680.19 rows=18844 width=8) (actual time=6.341..6.341 rows=19132 loops=1)
18	[...] Buckets: 32768 Batches: 1 Memory Usage: 1004kB
19	[...] -> Index Scan using grade_idx on "Workgroup" w (cost=0.42..680.19 rows=18844 width=8) (actual time=0.023..3.498 rows=19132 loops=1)
20	[...] Index Cond: (grade > '8'::numeric)
21	[...] -> Index Scan using "LabModule_pkey" on "LabModule" l (cost=0.42..839.47 rows=1 width=4) (actual time=2.982..3.275 rows=0 loops=789)
22	[...] Index Cond: (module_no = j.module_no)
23	[...] Filter: (type = 'project'::labmodule_type)
24	[...] Rows Removed by Filter: 1
25	Planning Time: 1.392 ms
26	Execution Time: 2696.882 ms

	QUERY PLAN	
	text	
1	Nested Loop (cost=2224.44..21167.06 rows=1 width=34) (actual time=14.260..2705.564 rows=172 loops=1)	
2	[...] Join Filter: (w.module_no = l.module_no)	
3	[...] -> Hash Join (cost=2224.03..20327.57 rows=1 width=42) (actual time=12.429..106.887 rows=789 loops=1)	
4	[...] Hash Cond: ((j."wgID" = w."wgID") AND (j.module_no = w.module_no))	
5	[...] -> Nested Loop (cost=1261.17..17263.52 rows=168095 width=42) (actual time=4.372..75.707 rows=132558 loops=1)	
6	[...] -> Hash Join (cost=1260.74..2761.95 rows=4381 width=58) (actual time=4.349..18.852 rows=4551 loops=1)	
7	[...] Hash Cond: ((p.amka)::text = (s.amka)::text)	
8	[...] -> Seq Scan on "Person" p (cost=0.00..1354.78 rows=55778 width=46) (actual time=0.033..3.337 rows=55778 loops=1)	
9	[...] -> Hash (cost=1205.98..1205.98 rows=4381 width=12) (actual time=4.271..4.271 rows=4551 loops=1)	
10	[...] Buckets: 8192 Batches: 1 Memory Usage: 260kB	
11	[...] -> Seq Scan on "Student" s (cost=0.00..1205.98 rows=4381 width=12) (actual time=0.025..3.791 rows=4551 loops=1)	
12	[...] Filter: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))	
13	[...] Rows Removed by Filter: 49381	
14	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..2.73 rows=58 width=20) (actual time=0.008..0.010 rows=29 loops=4551)	
15	[...] Index Cond: (amka = (p.amka)::text)	
16	[...] Heap Fetches: 0	
17	[...] -> Hash (cost=680.19..680.19 rows=18844 width=8) (actual time=7.888..7.888 rows=19132 loops=1)	
18	[...] Buckets: 32768 Batches: 1 Memory Usage: 1004kB	
19	[...] -> Index Scan using grade_idx on "Workgroup" w (cost=0.42..680.19 rows=18844 width=8) (actual time=0.023..4.491 rows=19132 loops=1)	
20	[...] Index Cond: (grade > '8'::numeric)	
21	[...] -> Index Scan using "LabModule_pkey" on "LabModule" l (cost=0.42..839.47 rows=1 width=4) (actual time=2.998..3.292 rows=0 loops=789)	
22	[...] Index Cond: (module_no = j.module_no)	
23	[...] Filter: (type = 'project'::labmodule_type)	
24	[...] Rows Removed by Filter: 1	
25	Planning Time: 1.365 ms	
26	Execution Time: 2705.798 ms	

Έπειτα, αξιοποιήσαμε την δυνατότητα ομαδοποίησης (clustering).

ΜΕ CLUSTER ΣΤΟ ΕΥΡΕΤΗΡΙΟ BTREE (grade)	
Εκτελέσεις	Χρόνος Απόκρισης (ms)
1 ^η	2477.643
2 ^η	2462.321
3 ^η	2496.148
ΜΟ	2478.704

Παρατηρήσαμε μικρή βελτίωση του χρόνου εκτέλεσης, καθώς με την συσταδοποίηση του Grade_idx ο χρόνος μειώθηκε περίπου κατά 8%, γεγονός αναμενόμενο από τη θεωρία.

	QUERY PLAN text
1	Nested Loop (cost=2243.43..22285.06 rows=1 width=34) (actual time=12.618..2477.349 rows=172 loops=1)
2	[...] Join Filter: (w.module_no = l.module_no)
3	[...] -> Hash Join (cost=2243.01..21445.57 rows=1 width=42) (actual time=10.566..95.327 rows=789 loops=1)
4	[...] Hash Cond: ((j."wgID" = w."wgID") AND (j.module_no = w.module_no))
5	[...] -> Nested Loop (cost=1264.80..18227.07 rows=179222 width=42) (actual time=4.442..69.766 rows=132558 loops=1)
6	[...] -> Hash Join (cost=1264.37..2765.58 rows=4671 width=58) (actual time=4.409..16.991 rows=4551 loops=1)
7	[...] Hash Cond: ((p.amka)::text = (s.amka)::text)
8	[...] -> Seq Scan on "Person" p (cost=0.00..1354.78 rows=55778 width=46) (actual time=0.019..3.078 rows=55778 loops=1)
9	[...] -> Hash (cost=1205.98..1205.98 rows=4671 width=12) (actual time=4.372..4.372 rows=4551 loops=1)
10	[...] Buckets: 8192 Batches: 1 Memory Usage: 260kB
11	[...] -> Seq Scan on "Student" s (cost=0.00..1205.98 rows=4671 width=12) (actual time=0.023..3.945 rows=4551 loops=1)
12	[...] Filter: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))
13	[...] Rows Removed by Filter: 49381
14	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..2.73 rows=58 width=20) (actual time=0.007..0.010 rows=29 loops=4551)

15	[...] Index Cond: (amka = (p.amka)::text)
16	[...] Heap Fetches: 0
17	[...] -> Hash (cost=691.23..691.23 rows=19132 width=8) (actual time=6.036..6.036 rows=19132 loops=1)
18	[...] Buckets: 32768 Batches: 1 Memory Usage: 1004kB
19	[...] -> Index Scan using grade_idx on "Workgroup" w (cost=0.42..691.23 rows=19132 width=8) (actual time=0.069..3.934 rows=19132 loops=1)
20	[...] Index Cond: (grade > '8'::numeric)
21	[...] -> Index Scan using "LabModule_pkey" on "LabModule" l (cost=0.42..839.47 rows=1 width=4) (actual time=2.748..3.018 rows=0 loops=789)
22	[...] Index Cond: (module_no = j.module_no)
23	[...] Filter: (type = 'project'::labmodule_type)
24	[...] Rows Removed by Filter: 1
25	Planning Time: 2.191 ms
26	Execution Time: 2477.643 ms

	QUERY PLAN text
1	Nested Loop (cost=2243.43..22285.06 rows=1 width=34) (actual time=12.259..2462.027 rows=172 loops=1)
2	[...] Join Filter: (w.module_no = l.module_no)
3	[...] -> Hash Join (cost=2243.01..21445.57 rows=1 width=42) (actual time=10.321..95.644 rows=789 loops=1)
4	[...] Hash Cond: ((j."wgID" = w."wgID") AND (j.module_no = w.module_no))
5	[...] -> Nested Loop (cost=1264.80..18227.07 rows=179222 width=42) (actual time=4.361..70.363 rows=132558 loops=1)
6	[...] -> Hash Join (cost=1264.37..2765.58 rows=4671 width=58) (actual time=4.334..17.082 rows=4551 loops=1)
7	[...] Hash Cond: ((p.amka)::text = (s.amka)::text)
8	[...] -> Seq Scan on "Person" p (cost=0.00..1354.78 rows=55778 width=46) (actual time=0.015..3.130 rows=55778 loops=1)
9	[...] -> Hash (cost=1205.98..1205.98 rows=4671 width=12) (actual time=4.308..4.308 rows=4551 loops=1)
10	[...] Buckets: 8192 Batches: 1 Memory Usage: 260kB
11	[...] -> Seq Scan on "Student" s (cost=0.00..1205.98 rows=4671 width=12) (actual time=0.019..3.896 rows=4551 loops=1)
12	[...] Filter: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))
13	[...] Rows Removed by Filter: 49381
14	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..2.73 rows=58 width=20) (actual time=0.007..0.010 rows=29 loops=4551)

15	[...] Index Cond: (amka = (p.amka)::text)
16	[...] Heap Fetches: 0
17	[...] -> Hash (cost=691.23..691.23 rows=19132 width=8) (actual time=5.880..5.880 rows=19132 loops=1)
18	[...] Buckets: 32768 Batches: 1 Memory Usage: 1004kB
19	[...] -> Index Scan using grade_idx on "Workgroup" w (cost=0.42..691.23 rows=19132 width=8) (actual time=0.120..3.897 rows=19132 loops=1)
20	[...] Index Cond: (grade > '8'::numeric)
21	[...] -> Index Scan using "LabModule_pkey" on "LabModule" l (cost=0.42..839.47 rows=1 width=4) (actual time=2.728..2.999 rows=0 loops=789)
22	[...] Index Cond: (module_no = j.module_no)
23	[...] Filter: (type = 'project'::labmodule_type)
24	[...] Rows Removed by Filter: 1
25	Planning Time: 2.005 ms
26	Execution Time: 2462.321 ms

QUERY PLAN		
	text	
1	Nested Loop (cost=2243.43..22285.06 rows=1 width=34) (actual time=12.566..2495.928 rows=172 loops=1)	
2	[...] Join Filter: (w.module_no = l.module_no)	
3	[...] -> Hash Join (cost=2243.01..21445.57 rows=1 width=42) (actual time=10.441..96.634 rows=789 loops=1)	
4	[...] Hash Cond: ((j."wgID" = w."wgID") AND (j.module_no = w.module_no))	
5	[...] -> Nested Loop (cost=1264.80..18227.07 rows=179222 width=42) (actual time=4.383..70.622 rows=132558 loops=1)	
6	[...] -> Hash Join (cost=1264.37..2765.58 rows=4671 width=58) (actual time=4.356..17.260 rows=4551 loops=1)	
7	[...] Hash Cond: ((p.amka)::text = (s.amka)::text)	
8	[...] -> Seq Scan on "Person" p (cost=0.00..1354.78 rows=55778 width=46) (actual time=0.014..3.156 rows=55778 loops=1)	
9	[...] -> Hash (cost=1205.98..1205.98 rows=4671 width=12) (actual time=4.306..4.306 rows=4551 loops=1)	
10	[...] Buckets: 8192 Batches: 1 Memory Usage: 260kB	
11	[...] -> Seq Scan on "Student" s (cost=0.00..1205.98 rows=4671 width=12) (actual time=0.020..3.830 rows=4551 loops=1)	
12	[...] Filter: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))	
13	[...] Rows Removed by Filter: 49381	
14	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..2.73 rows=58 width=20) (actual time=0.007..0.010 rows=29 loops=4551)	

15	[...] Index Cond: (amka = (p.amka)::text)
16	[...] Heap Fetches: 0
17	[...] -> Hash (cost=691.23..691.23 rows=19132 width=8) (actual time=5.882..5.882 rows=19132 loops=1)
18	[...] Buckets: 32768 Batches: 1 Memory Usage: 1004kB
19	[...] -> Index Scan using grade_idx on "Workgroup" w (cost=0.42..691.23 rows=19132 width=8) (actual time=0.067..3.603 rows=19132 loops=1)
20	[...] Index Cond: (grade > '8'::numeric)
21	[...] -> Index Scan using "LabModule_pkey" on "LabModule" l (cost=0.42..839.47 rows=1 width=4) (actual time=2.769..3.040 rows=0 loops=789)
22	[...] Index Cond: (module_no = j.module_no)
23	[...] Filter: (type = 'project'::labmodule_type)
24	[...] Rows Removed by Filter: 1
25	Planning Time: 1.866 ms
26	Execution Time: 2496.148 ms

Στη συνέχεια, έγινε απενεργοποίηση κάποιων αλγορίθμων υπολογισμού συνδέσεων τόσο πριν, όσο και μετά την δημιουργία των ευρετηρίων.

ΧΩΡΙΣ ΕΥΡΕΤΗΡΙΑ, ΧΩΡΙΣ hashjoin	
Εκτελέσεις	Χρόνος Απόκρισης (ms)
1 ^η	3540.134
2 ^η	3498.463
3 ^η	3464.515
ΜΟ	3501.037

Παρατηρούμε αύξηση του χρόνου εκτέλεσης κατά 4%, όπως και αναμέναμε εφόσον η χρήση του hashjoin βελτιώνει την απόδοση της εκτέλεσης.

QUERY PLAN		
	text	
1	Nested Loop (cost=55452.05..57787.96 rows=1 width=34) (actual time=455.624..3537.495 rows=172 loops=1)	
2	[...] Join Filter: (w.module_no = l.module_no)	
3	[...] -> Merge Join (cost=55451.63..56948.48 rows=1 width=42) (actual time=400.222..449.964 rows=789 loops=1)	
4	[...] Merge Cond: ((j.module_no = w.module_no) AND (j."wglD" = w."wglD"))	
5	[...] -> Sort (cost=42697.60..43150.26 rows=181064 width=42) (actual time=281.300..316.908 rows=132558 loops=1)	
6	[...] Sort Key: j.module_no, j."wglD"	
7	[...] Sort Method: external merge Disk: 6992kB	
8	[...] -> Nested Loop (cost=1494.87..21313.16 rows=181064 width=42) (actual time=21.379..157.019 rows=132558 loops=1)	
9	[...] -> Merge Join (cost=1494.44..6601.25 rows=4719 width=58) (actual time=21.335..88.291 rows=4551 loops=1)	
10	[...] Merge Cond: ((p.amka)::text = (s.amka)::text)	
11	[...] -> Index Scan using "Person_pkey" on "Person" p (cost=0.41..4897.08 rows=55778 width=46) (actual time=0.010..40.238 rows=55623 loops=1)	
12	[...] -> Sort (cost=1493.94..1505.74 rows=4719 width=12) (actual time=21.287..22.460 rows=4551 loops=1)	
13	[...] Sort Key: s.amka	
14	[...] Sort Method: quicksort Memory: 406kB	
15	[...] -> Seq Scan on "Student" s (cost=0.00..1205.98 rows=4719 width=12) (actual time=0.056..5.471 rows=4551 loops=1)	
16	[...] Filter: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))	
17	[...] Rows Removed by Filter: 49381	
18	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..2.53 rows=59 width=20) (actual time=0.009..0.013 rows=29 loops=4551)	
19	[...] Index Cond: (amka = (p.amka)::text)	
20	[...] Heap Fetches: 0	
21	[...] -> Sort (cost=12754.02..12800.31 rows=18518 width=8) (actual time=117.826..119.023 rows=12456 loops=1)	
22	[...] Sort Key: w.module_no, w."wglD"	
23	[...] Sort Method: quicksort Memory: 1665kB	
24	[...] -> Seq Scan on "Workgroup" w (cost=0.00..11441.40 rows=18518 width=8) (actual time=109.578..113.344 rows=19132 loops=1)	
25	[...] Filter: (grade > '8'::numeric)	
26	[...] Rows Removed by Filter: 557140	
27	[...] -> Index Scan using "LabModule_pkey" on "LabModule" l (cost=0.42..839.47 rows=1 width=4) (actual time=3.553..3.910 rows=0 loops=789)	
28	[...] Index Cond: (module_no = j.module_no)	
29	[...] Filter: (type = 'project'::labmodule_type)	
30	[...] Rows Removed by Filter: 1	
31	Planning Time: 1.899 ms	
32	Execution Time: 3540.134 ms	

	QUERY PLAN	
	text	
1	Nested Loop (cost=55452.05..57787.96 rows=1 width=34) (actual time=457.193..3495.751 rows=172 loops=1)	
2	[...] Join Filter: (w.module_no = l.module_no)	
3	[...] -> Merge Join (cost=55451.63..56948.48 rows=1 width=42) (actual time=402.496..452.185 rows=789 loops=1)	
4	[...] Merge Cond: ((j.module_no = w.module_no) AND (j."wglD" = w."wglD"))	
5	[...] -> Sort (cost=42697.60..43150.26 rows=181064 width=42) (actual time=286.195..322.085 rows=132558 loops=1)	
6	[...] Sort Key: j.module_no, j."wglD"	
7	[...] Sort Method: external merge Disk: 6992kB	
8	[...] -> Nested Loop (cost=1494.87..21313.16 rows=181064 width=42) (actual time=21.581..155.340 rows=132558 loops=1)	
9	[...] -> Merge Join (cost=1494.44..6601.25 rows=4719 width=58) (actual time=21.525..87.658 rows=4551 loops=1)	
10	[...] Merge Cond: ((p.amka)::text = (s.amka)::text)	
11	[...] -> Index Scan using "Person_pkey" on "Person" p (cost=0.41..4897.08 rows=55778 width=46) (actual time=0.024..39.953 rows=55623 loops=1)	
12	[...] -> Sort (cost=1493.94..1505.74 rows=4719 width=12) (actual time=21.465..22.646 rows=4551 loops=1)	
13	[...] Sort Key: s.amka	
14	[...] Sort Method: quicksort Memory: 406kB	
15	[...] -> Seq Scan on "Student" s (cost=0.00..1205.98 rows=4719 width=12) (actual time=0.048..5.349 rows=4551 loops=1)	
16	[...] Filter: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))	
17	[...] Rows Removed by Filter: 49381	
18	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..2.53 rows=59 width=20) (actual time=0.009..0.012 rows=29 loops=4551)	
19	[...] Index Cond: (amka = (p.amka)::text)	
20	[...] Heap Fetches: 0	
21	[...] -> Sort (cost=12754.02..12800.31 rows=18518 width=8) (actual time=115.148..116.309 rows=12456 loops=1)	
22	[...] Sort Key: w.module_no, w."wglD"	
23	[...] Sort Method: quicksort Memory: 1665kB	
24	[...] -> Seq Scan on "Workgroup" w (cost=0.00..11441.40 rows=18518 width=8) (actual time=105.933..109.952 rows=19132 loops=1)	
25	[...] Filter: (grade > '8'::numeric)	
26	[...] Rows Removed by Filter: 557140	
27	[...] -> Index Scan using "LabModule_pkey" on "LabModule" l (cost=0.42..839.47 rows=1 width=4) (actual time=3.494..3.855 rows=0 loops=789)	
28	[...] Index Cond: (module_no = j.module_no)	
29	[...] Filter: (type = 'project'::labmodule_type)	
30	[...] Rows Removed by Filter: 1	
31	Planning Time: 2.045 ms	
32	Execution Time: 3498.463 ms	

	QUERY PLAN	
	text	
1	Nested Loop (cost=55452.05..57787.96 rows=1 width=34) (actual time=451.139..3462.563 rows=172 loops=1)	
2	[...] Join Filter: (w.module_no = l.module_no)	
3	[...] -> Merge Join (cost=55451.63..56948.48 rows=1 width=42) (actual time=396.633..446.460 rows=789 loops=1)	
4	[...] Merge Cond: ((j.module_no = w.module_no) AND (j."wglD" = w."wglD"))	
5	[...] -> Sort (cost=42697.60..43150.26 rows=181064 width=42) (actual time=278.843..315.316 rows=132558 loops=1)	
6	[...] Sort Key: j.module_no, j."wglD"	
7	[...] Sort Method: external merge Disk: 6992kB	
8	[...] -> Nested Loop (cost=1494.87..21313.16 rows=181064 width=42) (actual time=21.574..155.774 rows=132558 loops=1)	
9	[...] -> Merge Join (cost=1494.44..6601.25 rows=4719 width=58) (actual time=21.531..88.265 rows=4551 loops=1)	
10	[...] Merge Cond: ((p.amka)::text = (s.amka)::text)	
11	[...] -> Index Scan using "Person_pkey" on "Person" p (cost=0.41..4897.08 rows=55778 width=46) (actual time=0.014..40.299 rows=55623 loops=1)	
12	[...] -> Sort (cost=1493.94..1505.74 rows=4719 width=12) (actual time=21.481..22.671 rows=4551 loops=1)	
13	[...] Sort Key: s.amka	
14	[...] Sort Method: quicksort Memory: 406kB	

15	[...] -> Seq Scan on "Student" s (cost=0.00..1205.98 rows=4719 width=12) (actual time=0.045..5.116 rows=4551 loops=1)
16	[...] Filter: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))
17	[...] Rows Removed by Filter: 49381
18	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..2.53 rows=59 width=20) (actual time=0.009..0.012 rows=29 loops=4551)
19	[...] Index Cond: (amka = (p.amka)::text)
20	[...] Heap Fetches: 0
21	[...] -> Sort (cost=12754.02..12800.31 rows=18518 width=8) (actual time=116.442..117.519 rows=12456 loops=1)
22	[...] Sort Key: w.module_no, w.wglID"
23	[...] Sort Method: quicksort Memory: 1665kB
24	[...] -> Seq Scan on "Workgroup" w (cost=0.00..11441.40 rows=18518 width=8) (actual time=107.248..111.345 rows=19132 loops=1)
25	[...] Filter: (grade > '8'::numeric)
26	[...] Rows Removed by Filter: 557140
27	[...] -> Index Scan using "LabModule_pkey" on "LabModule" l (cost=0.42..839.47 rows=1 width=4) (actual time=3.466..3.820 rows=0 loops=789)
28	[...] Index Cond: (module_no = j.module_no)
29	[...] Filter: (type = 'project'::labmodule_type)
30	[...] Rows Removed by Filter: 1
31	Planning Time: 1.727 ms
32	Execution Time: 3464.515 ms

ΧΩΡΙΣ ΕΥΡΕΤΗΡΙΑ, ΧΩΡΙΣ hashjoin, ΧΩΡΙΣ mergejoin	
Εκτελέσεις	Χρόνος Απόκρισης (ms)
1 ^η	76507.569
2 ^η	74507.439
3 ^η	77601.535
ΜΟ	76205.514

Μετά την απενεργοποίηση του mergejoin η απόδοση πέφτει ακόμα περισσότερο και συγκεκριμένα ο μέσος χρόνος εκτέλεσης αυξάνεται κατά 2164% σε σχέση με τον αρχικό. Το αποτέλεσμα αυτό είναι το αναμενόμενο.

	QUERY PLAN	
	text	
1	Nested Loop (cost=0.43..18363878.57 rows=1 width=34) (actual time=21542.284..76503.599 rows=172 loops=1)	
2	[...] Join Filter: ((s.amka)::text = (p.amka)::text)	
3	[...] Rows Removed by Join Filter: 4603280	
4	[...] -> Nested Loop (cost=0.43..18361826.56 rows=1 width=24) (actual time=21539.386..75896.117 rows=172 loops=1)	
5	[...] Join Filter: ((j."wglID" = w."wglID") AND (j.module_no = l.module_no))	
6	[...] Rows Removed by Join Filter: 511143476	
7	[...] -> Nested Loop (cost=0.43..27175.93 rows=181064 width=32) (actual time=0.142..102.296 rows=132558 loops=1)	
8	[...] -> Seq Scan on "Student" s (cost=0.00..1205.98 rows=4719 width=12) (actual time=0.068..6.936 rows=4551 loops=1)	
9	[...] Filter: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))	
10	[...] Rows Removed by Filter: 49381	
11	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..4.91 rows=59 width=20) (actual time=0.013..0.018 rows=29 loops=4551)	
12	[...] Index Cond: (amka = (s.amka)::text)	
13	[...] Heap Fetches: 0	
14	[...] -> Materialize (cost=0.00..6357276.49 rows=3780 width=12) (actual time=0.001..0.375 rows=3856 loops=132558)	

15	[...] -> Nested Loop (cost=0.00..6357257.58 rows=3780 width=12) (actual time=102.593..35387.699 rows=3856 loops=1)
16	[...] Join Filter: (w.module_no = l.module_no)
17	[...] Rows Removed by Join Filter: 428189436
18	[...] -> Seq Scan on "LabModule" l (cost=0.00..2336.40 rows=22837 width=4) (actual time=0.013..17.740 rows=22381 loops=1)
19	[...] Filter: (type = 'project'::labmodule_type)
20	[...] Rows Removed by Filter: 89491
21	[...] -> Materialize (cost=0.00..11533.99 rows=18518 width=8) (actual time=0.005..0.556 rows=19132 loops=22381)
22	[...] -> Seq Scan on "Workgroup" w (cost=0.00..11441.40 rows=18518 width=8) (actual time=102.570..106.523 rows=19132 loops=1)
23	[...] Filter: (grade > '8'::numeric)
24	[...] Rows Removed by Filter: 557140
25	[...] -> Seq Scan on "Person" p (cost=0.00..1354.78 rows=55778 width=46) (actual time=0.002..1.322 rows=26764 loops=172)
26	Planning Time: 1.300 ms
27	Execution Time: 76507.569 ms

	QUERY PLAN	
	text	
1	Nested Loop (cost=0.43..17918210.66 rows=1 width=34) (actual time=19897.659..74507.046 rows=172 loops=1)	
2	[...] Join Filter: ((s.amka)::text = (p.amka)::text)	
3	[...] Rows Removed by Join Filter: 4603280	
4	[...] -> Nested Loop (cost=0.43..17916158.65 rows=1 width=24) (actual time=19895.505..73927.355 rows=172 loops=1)	
5	[...] Join Filter: ((j."wglD" = w."wglD") AND (j.module_no = l.module_no))	
6	[...] Rows Removed by Join Filter: 511143476	
7	[...] -> Nested Loop (cost=0.43..26281.08 rows=174273 width=32) (actual time=0.114..96.734 rows=132558 loops=1)	
8	[...] -> Seq Scan on "Student" s (cost=0.00..1205.98 rows=4542 width=12) (actual time=0.019..6.519 rows=4551 loops=1)	
9	[...] Filter: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))	
10	[...] Rows Removed by Filter: 49381	
11	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..4.93 rows=59 width=20) (actual time=0.013..0.017 rows=29 loops=4551)	
12	[...] Index Cond: (amka = (s.amka)::text)	
13	[...] Heap Fetches: 0	
14	[...] -> Materialize (cost=0.00..6358678.30 rows=3781 width=12) (actual time=0.001..0.361 rows=3856 loops=132558)	
15	[...] -> Nested Loop (cost=0.00..6358659.40 rows=3781 width=12) (actual time=82.538..33709.189 rows=3856 loops=1)	
16	[...] Join Filter: (w.module_no = l.module_no)	
17	[...] Rows Removed by Join Filter: 428189436	
18	[...] -> Seq Scan on "LabModule" l (cost=0.00..2336.40 rows=22378 width=4) (actual time=0.004..13.026 rows=22381 loops=1)	
19	[...] Filter: (type = 'project'::labmodule_type)	
20	[...] Rows Removed by Filter: 89491	
21	[...] -> Materialize (cost=0.00..11535.91 rows=18902 width=8) (actual time=0.004..0.527 rows=19132 loops=22381)	
22	[...] -> Seq Scan on "Workgroup" w (cost=0.00..11441.40 rows=18902 width=8) (actual time=82.528..85.785 rows=19132 loops=1)	
23	[...] Filter: (grade > '8'::numeric)	
24	[...] Rows Removed by Filter: 557140	
25	[...] -> Seq Scan on "Person" p (cost=0.00..1354.78 rows=55778 width=46) (actual time=0.002..1.289 rows=26764 loops=172)	
26	Planning Time: 1.145 ms	
27	Execution Time: 74507.439 ms	

	QUERY PLAN text	
1	Nested Loop (cost=0.43..17918210.66 rows=1 width=34) (actual time=19966.984..77601.197 rows=172 loops=1)	
2	[...] Join Filter: ((s.amka)::text = (p.amka)::text)	
3	[...] Rows Removed by Join Filter: 4603280	
4	[...] -> Nested Loop (cost=0.43..17916158.65 rows=1 width=24) (actual time=19964.787..77017.347 rows=172 loops=1)	
5	[...] Join Filter: ((j."wgID" = w."wgID") AND (j.module_no = l.module_no))	
6	[...] Rows Removed by Join Filter: 511143476	
7	[...] -> Nested Loop (cost=0.43..26281.08 rows=174273 width=32) (actual time=0.059..116.105 rows=132558 loops=1)	
8	[...] -> Seq Scan on "Student" s (cost=0.00..1205.98 rows=4542 width=12) (actual time=0.025..9.448 rows=4551 loops=1)	
9	[...] Filter: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))	
10	[...] Rows Removed by Filter: 49381	
11	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..4.93 rows=59 width=20) (actual time=0.015..0.020 rows=29 loops=4551)	
12	[...] Index Cond: (amka = (s.amka)::text)	
13	[...] Heap Fetches: 0	
14	[...] -> Materialize (cost=0.00..6358678.30 rows=3781 width=12) (actual time=0.001..0.383 rows=3856 loops=132558)	
15	[...] -> Nested Loop (cost=0.00..6358659.40 rows=3781 width=12) (actual time=83.427..33770.320 rows=3856 loops=1)	
16	[...] Join Filter: (w.module_no = l.module_no)	
17	[...] Rows Removed by Join Filter: 428189436	
18	[...] -> Seq Scan on "LabModule" l (cost=0.00..2336.40 rows=22378 width=4) (actual time=0.005..17.191 rows=22381 loops=1)	
19	[...] Filter: (type = 'project'::labmodule_type)	
20	[...] Rows Removed by Filter: 89491	
21	[...] -> Materialize (cost=0.00..11535.91 rows=18902 width=8) (actual time=0.004..0.529 rows=19132 loops=22381)	
22	[...] -> Seq Scan on "Workgroup" w (cost=0.00..11441.40 rows=18902 width=8) (actual time=83.415..86.701 rows=19132 loops=1)	
23	[...] Filter: (grade > '8'::numeric)	
24	[...] Rows Removed by Filter: 557140	
25	[...] -> Seq Scan on "Person" p (cost=0.00..1354.78 rows=55778 width=46) (actual time=0.002..1.301 rows=26764 loops=172)	
26	Planning Time: 0.627 ms	
27	Execution Time: 77601.535 ms	

ME ΕΥΡΕΤΗΡΙΑ, ΧΩΡΙΣ hashjoin	
Εκτελέσεις	Χρόνος Απόκρισης (ms)
1 ^η	2855.602
2 ^η	2817.789
3 ^η	2837.171
ΜΟ	2855.602

Όπως και πριν, έχουμε αύξηση του χρόνου εκτέλεσης, αυτή τη φορά όμως κατά 6% σε σχέση με την εκτέλεση χρήση ευρετηρίου και hashjoin.

	QUERY PLAN text
1	Nested Loop (cost=43989.96..46272.49 rows=1 width=34) (actual time=249.119..2854.135 rows=172 loops=1)
2	[...] Join Filter: (w.module_no = l.module_no)
3	[...] -> Merge Join (cost=43989.54..45433.01 rows=1 width=42) (actual time=200.646..237.372 rows=789 loops=1)
4	[...] Merge Cond: ((j.module_no = w.module_no) AND (j."wglD" = w."wglD"))
5	[...] -> Sort (cost=41916.07..42348.97 rows=173160 width=42) (actual time=191.517..218.703 rows=132558 loops=1)
6	[...] Sort Key: j.module_no, j."wglD"
7	[...] Sort Method: external merge Disk: 6992kB
8	[...] -> Nested Loop (cost=1480.76..21522.63 rows=173160 width=42) (actual time=16.405..112.132 rows=132558 loops=1)
9	[...] -> Merge Join (cost=1480.33..6584.13 rows=4513 width=58) (actual time=16.383..61.444 rows=4551 loops=1)
10	[...] Merge Cond: ((p.amka)::text = (s.amka)::text)
11	[...] -> Index Scan using "Person_pkey" on "Person" p (cost=0.41..4897.08 rows=55778 width=46) (actual time=0.009..26.419 rows=55623 loops=1)
12	[...] -> Sort (cost=1479.92..1491.20 rows=4513 width=12) (actual time=16.354..17.017 rows=4551 loops=1)
13	[...] Sort Key: s.amka
14	[...] Sort Method: quicksort Memory: 406kB

15	[...] -> Seq Scan on "Student" s (cost=0.00..1205.98 rows=4513 width=12) (actual time=0.025..4.692 rows=4551 loops=1)
16	[...] Filter: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))
17	[...] Rows Removed by Filter: 49381
18	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..2.73 rows=58 width=20) (actual time=0.006..0.009 rows=29 loops=4551)
19	[...] Index Cond: (amka = (p.amka)::text)
20	[...] Heap Fetches: 0
21	[...] -> Sort (cost=2073.46..2121.72 rows=19305 width=8) (actual time=8.317..9.085 rows=12456 loops=1)
22	[...] Sort Key: w.module_no, w."wglD"
23	[...] Sort Method: quicksort Memory: 1665kB
24	[...] -> Index Scan using grade_idx on "Workgroup" w (cost=0.42..699.26 rows=19305 width=8) (actual time=0.033..3.555 rows=19132 loops=1)
25	[...] Index Cond: (grade > '8'::numeric)
26	[...] -> Index Scan using "LabModule_pkey" on "LabModule" l (cost=0.42..839.47 rows=1 width=4) (actual time=3.012..3.315 rows=0 loops=789)
27	[...] Index Cond: (module_no = j.module_no)
28	[...] Filter: (type = 'project'::labmodule_type)
29	[...] Rows Removed by Filter: 1
30	Planning Time: 1.438 ms
31	Execution Time: 2855.602 ms

	QUERY PLAN text
1	Nested Loop (cost=43989.96..46272.49 rows=1 width=34) (actual time=240.963..2816.367 rows=172 loops=1)
2	[...] Join Filter: (w.module_no = l.module_no)
3	[...] -> Merge Join (cost=43989.54..45433.01 rows=1 width=42) (actual time=194.493..230.527 rows=789 loops=1)
4	[...] Merge Cond: ((j.module_no = w.module_no) AND (j."wglD" = w."wglD"))
5	[...] -> Sort (cost=41916.07..42348.97 rows=173160 width=42) (actual time=186.037..213.094 rows=132558 loops=1)
6	[...] Sort Key: j.module_no, j."wglD"
7	[...] Sort Method: external merge Disk: 6992kB
8	[...] -> Nested Loop (cost=1480.76..21522.63 rows=173160 width=42) (actual time=15.576..106.947 rows=132558 loops=1)
9	[...] -> Merge Join (cost=1480.33..6584.13 rows=4513 width=58) (actual time=15.556..58.898 rows=4551 loops=1)
10	[...] Merge Cond: ((p.amka)::text = (s.amka)::text)
11	[...] -> Index Scan using "Person_pkey" on "Person" p (cost=0.41..4897.08 rows=55778 width=46) (actual time=0.009..25.628 rows=55623 loops=1)
12	[...] -> Sort (cost=1479.92..1491.20 rows=4513 width=12) (actual time=15.534..16.129 rows=4551 loops=1)
13	[...] Sort Key: s.amka
14	[...] Sort Method: quicksort Memory: 406kB

15	[...] -> Seq Scan on "Student" s (cost=0.00..1205.98 rows=4513 width=12) (actual time=0.025..4.042 rows=4551 loops=1)
16	[...] Filter: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))
17	[...] Rows Removed by Filter: 49381
18	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..2.73 rows=58 width=20) (actual time=0.006..0.009 rows=29 loops=4551)
19	[...] Index Cond: (amka = (p.amka)::text)
20	[...] Heap Fetches: 0
21	[...] -> Sort (cost=2073.46..2121.72 rows=19305 width=8) (actual time=7.663..8.381 rows=12456 loops=1)
22	[...] Sort Key: w.module_no, w."wglD"
23	[...] Sort Method: quicksort Memory: 1665kB
24	[...] -> Index Scan using grade_idx on "Workgroup" w (cost=0.42..699.26 rows=19305 width=8) (actual time=0.034..3.202 rows=19132 loops=1)
25	[...] Index Cond: (grade > '8'::numeric)
26	[...] -> Index Scan using "LabModule_pkey" on "LabModule" l (cost=0.42..839.47 rows=1 width=4) (actual time=2.985..3.276 rows=0 loops=789)
27	[...] Index Cond: (module_no = j.module_no)
28	[...] Filter: (type = 'project'::labmodule_type)
29	[...] Rows Removed by Filter: 1
30	Planning Time: 1.355 ms
31	Execution Time: 2817.789 ms

	QUERY PLAN	
	text	
1	Nested Loop (cost=43989.96..46272.49 rows=1 width=34) (actual time=261.959..2835.790 rows=172 loops=1)	
2	[...] Join Filter: (w.module_no = l.module_no)	
3	[...] -> Merge Join (cost=43989.54..45433.01 rows=1 width=42) (actual time=216.139..254.352 rows=789 loops=1)	
4	[...] Merge Cond: ((j.module_no = w.module_no) AND (j."wglD" = w."wglD"))	
5	[...] -> Sort (cost=41916.07..42348.97 rows=173160 width=42) (actual time=207.388..235.462 rows=132558 loops=1)	
6	[...] Sort Key: j.module_no, j."wglD"	
7	[...] Sort Method: external merge Disk: 6992kB	
8	[...] -> Nested Loop (cost=1480.76..21522.63 rows=173160 width=42) (actual time=18.389..118.023 rows=132558 loops=1)	
9	[...] -> Merge Join (cost=1480.33..6584.13 rows=4513 width=58) (actual time=18.366..64.969 rows=4551 loops=1)	
10	[...] Merge Cond: ((p.amka)::text = (s.amka)::text)	
11	[...] -> Index Scan using "Person_pkey" on "Person" p (cost=0.41..4897.08 rows=55778 width=46) (actual time=0.010..27.359 rows=55623 loops=1)	
12	[...] -> Sort (cost=1479.92..1491.20 rows=4513 width=12) (actual time=18.340..18.944 rows=4551 loops=1)	
13	[...] Sort Key: s.amka	
14	[...] Sort Method: quicksort Memory: 406kB	

15	[...] -> Seq Scan on "Student" s (cost=0.00..1205.98 rows=4513 width=12) (actual time=0.024..6.047 rows=4551 loops=1)
16	[...] Filter: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))
17	[...] Rows Removed by Filter: 49381
18	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..2.73 rows=58 width=20) (actual time=0.007..0.010 rows=29 loops=4551)
19	[...] Index Cond: (amka = (p.amka)::text)
20	[...] Heap Fetches: 0
21	[...] -> Sort (cost=2073.46..2121.72 rows=19305 width=8) (actual time=7.945..8.654 rows=12456 loops=1)
22	[...] Sort Key: w.module_no, w."wglD"
23	[...] Sort Method: quicksort Memory: 1665kB
24	[...] -> Index Scan using grade_idx on "Workgroup" w (cost=0.42..699.26 rows=19305 width=8) (actual time=0.037..3.315 rows=19132 loops=1)
25	[...] Index Cond: (grade > '8'::numeric)
26	[...] -> Index Scan using "LabModule_pkey" on "LabModule" l (cost=0.42..839.47 rows=1 width=4) (actual time=2.983..3.270 rows=0 loops=789)
27	[...] Index Cond: (module_no = j.module_no)
28	[...] Filter: (type = 'project'::labmodule_type)
29	[...] Rows Removed by Filter: 1
30	Planning Time: 1.442 ms
31	Execution Time: 2837.171 ms

ΜΕ ΕΥΡΕΤΗΡΙΑ, ΧΩΡΙΣ hashjoin, ΧΩΡΙΣ mergejoin	
Εκτελέσεις	Χρόνος Απόκρισης (ms)
1 ^η	71685.511
2 ^η	73220.403
3 ^η	73707.551
ΜΟ	72871.155

Εδώ ο χρόνος εκτέλεσης αυξάνεται πολύ περισσότερο και συγκεκριμένα κατά 2597% σε σχέση με την εκτέλεση με χρήση ευρετηρίων και τα enable_hashjoin και enable_mergejoin on.

	QUERY PLAN	
	text	
1	Nested Loop (cost=0.85..17436228.59 rows=1 width=34) (actual time=19864.346..71685.261 rows=172 loops=1)	
2	[...] Join Filter: ((s.amka)::text = (p.amka)::text)	
3	[...] Rows Removed by Join Filter: 4603280	
4	[...] -> Nested Loop (cost=0.85..17434176.58 rows=1 width=24) (actual time=19862.373..71162.224 rows=172 loops=1)	
5	[...] Join Filter: ((j."wgID" = w."wgID") AND (j.module_no = l.module_no))	
6	[...] Rows Removed by Join Filter: 511143476	
7	[...] -> Nested Loop (cost=0.43..26506.11 rows=174157 width=32) (actual time=0.062..117.394 rows=132558 loops=1)	
8	[...] -> Seq Scan on "Student" s (cost=0.00..1205.98 rows=4539 width=12) (actual time=0.028..7.329 rows=4551 loops=1)	
9	[...] Filter: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))	
10	[...] Rows Removed by Filter: 49381	
11	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..4.99 rows=58 width=20) (actual time=0.013..0.020 rows=29 loops=4551)	
12	[...] Index Cond: (amka = (s.amka)::text)	
13	[...] Heap Fetches: 0	
14	[...] -> Materialize (cost=0.42..6182825.64 rows=3683 width=12) (actual time=0.000..0.352 rows=3856 loops=132558)	
15	[...] -> Nested Loop (cost=0.42..6182807.23 rows=3683 width=12) (actual time=0.037..32864.412 rows=3856 loops=1)	
16	[...] Join Filter: (w.module_no = l.module_no)	
17	[...] Rows Removed by Join Filter: 428189436	
18	[...] -> Seq Scan on "LabModule" l (cost=0.00..2336.40 rows=21942 width=4) (actual time=0.009..25.784 rows=22381 loops=1)	
19	[...] Filter: (type = 'project'::labmodule_type)	
20	[...] Rows Removed by Filter: 89491	
21	[...] -> Materialize (cost=0.42..772.88 rows=18776 width=8) (actual time=0.000..0.539 rows=19132 loops=22381)	
22	[...] -> Index Scan using grade_idx on "Workgroup" w (cost=0.42..679.00 rows=18776 width=8) (actual time=0.025..3.575 rows=19132 loops=1)	
23	[...] Index Cond: (grade > '8'::numeric)	
24	[...] -> Seq Scan on "Person" p (cost=0.00..1354.78 rows=55778 width=46) (actual time=0.002..1.282 rows=26764 loops=172)	
25	Planning Time: 0.644 ms	
26	Execution Time: 71685.511 ms	

	QUERY PLAN text
1	Nested Loop (cost=0.85..17436228.59 rows=1 width=34) (actual time=19028.750..73220.157 rows=172 loops=1)
2	[...] Join Filter: ((s.amka)::text = (p.amka)::text)
3	[...] Rows Removed by Join Filter: 4603280
4	[...] -> Nested Loop (cost=0.85..17434176.58 rows=1 width=24) (actual time=19026.835..72682.518 rows=172 loops=1)
5	[...] Join Filter: ((j."wglID" = w."wglID") AND (j.module_no = l.module_no))
6	[...] Rows Removed by Join Filter: 511143476
7	[...] -> Nested Loop (cost=0.43..26506.11 rows=174157 width=32) (actual time=0.060..135.018 rows=132558 loops=1)
8	[...] -> Seq Scan on "Student" s (cost=0.00..1205.98 rows=4539 width=12) (actual time=0.025..8.676 rows=4551 loops=1)
9	[...] Filter: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))
10	[...] Rows Removed by Filter: 49381
11	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..4.99 rows=58 width=20) (actual time=0.015..0.023 rows=29 loops=4551)
12	[...] Index Cond: (amka = (s.amka)::text)
13	[...] Heap Fetches: 0
14	[...] -> Materialize (cost=0.42..6182825.64 rows=3683 width=12) (actual time=0.000..0.349 rows=3856 loops=132558)
15	[...] -> Nested Loop (cost=0.42..6182807.23 rows=3683 width=12) (actual time=0.034..32367.924 rows=3856 loops=1)
16	[...] Join Filter: (w.module_no = l.module_no)
17	[...] Rows Removed by Join Filter: 428189436
18	[...] -> Seq Scan on "LabModule" l (cost=0.00..2336.40 rows=21942 width=4) (actual time=0.009..16.928 rows=22381 loops=1)
19	[...] Filter: (type = 'project'::labmodule_type)
20	[...] Rows Removed by Filter: 89491
21	[...] -> Materialize (cost=0.42..772.88 rows=18776 width=8) (actual time=0.000..0.524 rows=19132 loops=22381)
22	[...] -> Index Scan using grade_idx on "Workgroup" w (cost=0.42..679.00 rows=18776 width=8) (actual time=0.022..3.406 rows=19132 loops=1)
23	[...] Index Cond: (grade > '8'::numeric)
24	[...] -> Seq Scan on "Person" p (cost=0.00..1354.78 rows=55778 width=46) (actual time=0.002..1.299 rows=26764 loops=172)
25	Planning Time: 0.610 ms
26	Execution Time: 73220.403 ms

	QUERY PLAN text
1	Nested Loop (cost=0.85..17436228.59 rows=1 width=34) (actual time=19167.932..73707.184 rows=172 loops=1)
2	[...] Join Filter: ((s.amka)::text = (p.amka)::text)
3	[...] Rows Removed by Join Filter: 4603280
4	[...] -> Nested Loop (cost=0.85..17434176.58 rows=1 width=24) (actual time=19166.014..73171.014 rows=172 loops=1)
5	[...] Join Filter: ((j."wglID" = w."wglID") AND (j.module_no = l.module_no))
6	[...] Rows Removed by Join Filter: 511143476
7	[...] -> Nested Loop (cost=0.43..26506.11 rows=174157 width=32) (actual time=0.054..132.887 rows=132558 loops=1)
8	[...] -> Seq Scan on "Student" s (cost=0.00..1205.98 rows=4539 width=12) (actual time=0.028..11.927 rows=4551 loops=1)
9	[...] Filter: ((entry_date > '2010-09-01'::date) AND (entry_date < '2011-09-01'::date))
10	[...] Rows Removed by Filter: 49381
11	[...] -> Index Only Scan using "Joins_pkey" on "Joins" j (cost=0.43..4.99 rows=58 width=20) (actual time=0.014..0.021 rows=29 loops=4551)
12	[...] Index Cond: (amka = (s.amka)::text)
13	[...] Heap Fetches: 0
14	[...] -> Materialize (cost=0.42..6182825.64 rows=3683 width=12) (actual time=0.000..0.356 rows=3856 loops=132558)

15	[...] -> Nested Loop (cost=0.42..6182807.23 rows=3683 width=12) (actual time=0.033..33326.390 rows=3856 loops=1)
16	[...] Join Filter: (w.module_no = l.module_no)
17	[...] Rows Removed by Join Filter: 428189436
18	[...] -> Seq Scan on "LabModule" l (cost=0.00..2336.40 rows=21942 width=4) (actual time=0.009..17.184 rows=22381 loops=1)
19	[...] Filter: (type = 'project'::labmodule_type)
20	[...] Rows Removed by Filter: 89491
21	[...] -> Materialize (cost=0.42..772.88 rows=18776 width=8) (actual time=0.000..0.527 rows=19132 loops=22381)
22	[...] -> Index Scan using grade_idx on "Workgroup" w (cost=0.42..679.00 rows=18776 width=8) (actual time=0.022..3.345 rows=19132 loops=1)
23	[...] Index Cond: (grade > '8'::numeric)
24	[...] -> Seq Scan on "Person" p (cost=0.00..1354.78 rows=55778 width=46) (actual time=0.002..1.288 rows=26764 loops=172)
25	Planning Time: 0.591 ms
26	Execution Time: 73707.551 ms

Τέλος, εναλλάσσοντας την σειρά των συνδέσεων (join) δεν παρατηρήσαμε κάποια αλλαγή στην απόδοση. Οπότε, δεν το συμπεριλάβαμε στην αναφορά.

Γενικές Παρατηρήσεις:

- Η δημιουργία των κατάλληλων ευρετηρίων και η ομαδοποίηση με βάση αυτά έχουν ως αποτέλεσμα σημαντική αύξηση της αποδοτικότητας και μείωση του χρόνου εκτέλεσης ενός συγκεκριμένου query.
- Ωστόσο, δεν είναι όλα τα ευρετήρια κατάλληλα για την εκτέλεση ενός query, καθώς δεν βελτιώνουν πάντα τον χρόνο εκτέλεσης. Σε αυτές τις περιπτώσεις προτιμάμε να μην τα χρησιμοποιούμε, καθώς επιβαρύνουν το σύστημα άσκοπα.
- Επιπλέον, χωρίς τους αλγορίθμους υπολογισμού συνδέσεων η εκτέλεση του query επιβραδύνεται σημαντικά, με αποτέλεσμα η απόδοση να πέφτει σε μεγάλο βαθμό.
- Είναι σημαντικό να διερευνάται ο καλύτερος πιθανός τρόπος συνδέσεων (join), καθώς ακόμα και αυτό μπορεί να επιταχύνει ή να επιβραδύνει την εκτέλεση ενός query.